

Growth of the Complete Original Gamma Return

Finite positive bounds, exact equilibrium and arithmetic transport

Split-Zero research programme

14 September 2026

This paper continues the complete low-endpoint, parity and positive-shift calculation. It proves finite positive bounds for the unchanged Gamma functional, evaluates its leading coefficient by a full equilibrium proof, and returns that value through the original arithmetic and mixed-state maps. The full previous low and high source definitions and exact positive recurrence are printed first. All prerequisite proof sources accompany the editable paper; historical editions remain preserved.

Original source, low endpoint and exact finite recurrence

This calculation continues the full-source comparison in BRD1–42 and CTR1–54. It treats the original one-column relation determinant separately, so the degree condition $r \geq l$ of the projector construction is never applied to the low endpoint. Every comparison below is on the original scalar coefficient space, with the original polynomial multiplication maps and all four tail integrals present. The complete BRD, FTC and CTR sources supply the proved contour, polynomial and high-endpoint formulas.

1 Objects and the low source maps

Retain

$$k = 4l + 1, \quad l \geq 1, \quad m \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad 0 < \delta < 1/2, \quad \gamma > 2. \quad (\text{LET1})$$

Set $d = \delta^2$, $g = \gamma^2$, and define the complete original objects

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \\ f(S) &= \prod_{j=0}^{l-1} (S - c - t_j), \quad t_j = 2j + \frac{1}{2}, \quad \sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}. \end{aligned} \quad (\text{LET2})$$

Write $m_j = \int_{\mathbb{R}} y^j \sigma(y) dy$, with the index denoting the literal power of y . Thus $m_0 = \sqrt{2\pi}$, odd moments vanish, and CTR50–53 supply all even moments by their exact rational recurrence with the full mass factor. The original low relation endpoint is

$$\mathfrak{r}_\chi = \frac{\int_{\mathbb{R}} |f(c + iy)\chi(c + iy)|^2 \sigma(y) dy}{\int_{\mathbb{R}} |\chi(c + iy)|^2 \sigma(y) dy}. \quad (\text{LET3})$$

All four scalar integrals used below are finite by the proved Gamma upper bound and strictly positive because their polynomial factors vanish at only finitely many real points.

Put $\epsilon = 2^{-10}$, and partition the line into $I = \{|y| < q\epsilon\}$, $B = \{q\epsilon \leq |y| \leq 64q\}$, and $F = \{|y| > 64q\}$. Assume the explicit finite threshold

$$k \geq 2048\sqrt{d+g}. \quad (\text{LET4})$$

Then $q \geq k^2$ gives $k\sqrt{d+g}/q \leq \epsilon/2$. Also $t_{l-1} < k/2$ and $\sqrt{d+g} > 2$ give $t_{l-1}/q \leq \epsilon$. This is the same original-parameter domain already used for the complete high-endpoint comparison.

On the bulk Hilbert space $L^2(B, \sigma(y) dy)$, the maps from the one-dimensional coefficient space are

$$T_{\text{pow}}u(y) = (iy)^q u, \quad T_\chi u(y) = \chi(c + iy)u, \quad (Uh)(y) = \frac{\chi(c + iy)}{(iy)^q} h(y). \quad (\text{LET5})$$

Their exact identities are $UT_{\text{pow}} = T_\chi$ and $UM_f = M_f U$, where M_f is multiplication by $f(c + iy)$. Both U and its inverse are bounded: the bulk is compact, $y \neq 0$ there, and LET4 places every displacement root strictly inside its inner radius. The squared modulus of U is the original density ratio. Taking scalar Hilbert Grams proves the source comparisons below while retaining the original arithmetic quotient and its multiplication maps.

2 A scalar ratio retaining the degree l

Let w be a finite positive measure, $F_0 \geq 0$ a measurable function with $0 < \int F_0 dw < \infty$, and let ψ be a bounded real function. Define the two literal ratios

$$R_w = \frac{\int F_0 dw}{\int dw}, \quad R_{e^\psi w} = \frac{\int F_0 e^\psi dw}{\int e^\psi dw}.$$

If $a \leq \psi \leq b$ almost everywhere, positivity of the integrals gives

$$e^a \leq \frac{\int F_0 e^\psi dw}{\int F_0 dw} \leq e^b, \quad e^a \leq \frac{\int e^\psi dw}{\int dw} \leq e^b.$$

Division and logarithms therefore prove the exact finite estimate

$$|\log R_{e^\psi w} - \log R_w| \leq b - a. \quad (\text{LET6})$$

No restriction on the degree of $F_0 = |f|^2$ enters this scalar proof.

Apply LET6 to $dw = 1_B(y)|y|^{2q}\sigma(y) dy$ and $F_0(y) = |f(c + iy)|^2$. For $y = qz$, the unchanged root product has the expansion

$$\psi_\chi(qz) := \log \frac{|\chi(c + iqz)|^2}{|qz|^{2q}} = \frac{a_k}{z^2} + E_\chi(qz), \quad a_k = \frac{k(k+2)(d-g)}{3q}, \quad |E_\chi(qz)| \leq \delta_\chi, \\ \tau_\epsilon = \frac{k^2(d+g)}{q^2\epsilon^2} \leq \frac{1}{4}, \quad \delta_\chi = \frac{q\tau_\epsilon^2}{2(1-\tau_\epsilon)}. \quad (\text{LET7})$$

For completeness, write each factor as $iqz(1 + i\zeta_{ab}/(qz))$. The logarithmic Taylor series converges absolutely; the permutation $(a, b) \mapsto (k-a, k-b)$ cancels odd powers. The quadratic sum is $e \operatorname{Re} \sum_{a,b} \zeta_{ab}^2 = qk(k+2)(d-g)/3$. The remaining even powers have absolute sum at most $q \sum_{j \geq 2} \tau_\epsilon^j/j \leq q\tau_\epsilon^2/[2(1-\tau_\epsilon)]$. This proves every coefficient and remainder in LET7.

Since $z^{-2} \in [64^{-2}, \epsilon^{-2}]$, define the finite constant

$$E_0 = |a_k|(\epsilon^{-2} - 64^{-2}) + 2\delta_\chi. \quad (\text{LET8})$$

The exact bulk difference

$$d_\chi = \log \frac{\int_B |f\chi|^2 d\sigma}{\int_B |\chi|^2 d\sigma} - \log \frac{\int_B |f(c+iy)|^2 |y|^{2q} d\sigma}{\int_B |y|^{2q} d\sigma}$$

satisfies $|d_\chi| \leq E_0$ by LET6. All products in the first ratio are evaluated at $c+iy$. The original numerator multiplication also gives

$$1 \leq \frac{|f(c+iy)|^2}{|y|^{2l}} = \prod_{j=0}^{l-1} (1 + t_j^2/y^2) \leq e^{\Delta_f}, \quad \Delta_f = \frac{S_2}{q^2 \epsilon^2}, \quad S_2 = \sum_{j=0}^{l-1} t_j^2 = \frac{l(16l^2 - 12l - 1)}{12}. \quad (\text{LET9})$$

The upper bound follows by integrating the inequality $(\log(1+x))' = 1/(1+x) \leq 1$ from zero. Expanding each t_j^2 and summing j, j^2 gives the stated S_2 . Consequently the logarithmic correction d_f obtained by replacing the bulk numerator $|f(c+iy)|^2 |y|^{2q}$ by $|y|^{2q+2l}$ satisfies $0 \leq d_f \leq \Delta_f$, with that sign.

3 The four scalar tails on the original line

Consider the four positive integrals with weights

$$h_{\chi,0} = |\chi(c+iy)|^2, \quad h_{\chi,f} = |f(c+iy)\chi(c+iy)|^2, \quad h_{\text{pow},0} = |y|^{2q}, \quad h_{\text{pow},l} = |y|^{2q+2l}.$$

For each weight let $H = \int h d\sigma = H^I + H^B + H^F$ and

$$t_H = \log \left(1 + \frac{H^I + H^F}{H^B} \right) \geq 0, \quad \log H = \log H^B + t_H. \quad (\text{LET10})$$

Here every $H^B > 0$. We prove a common bound at scalar rank one rather than applying a high-rank statement outside its specified domain.

The original contour and reflection calculation BRD23–24 proves

$$C_L e^{-\pi|y|} \leq \sigma(y) \leq C_U e^{-|y|}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad C_U = \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos 1}}, \quad C_U/C_L = (\cos 1)^{-1/2}. \quad (\text{LET11})$$

Those proofs retain the Euler contour rotation and keyhole residue; they are included in the complete source dependencies. Pairing the roots of χ as $\zeta, -\zeta$ gives

$$|\chi(c+iy)|^2 \leq (y^2 + k^2(d+g))^q.$$

On $1 \leq z \leq 2$, the reverse triangle inequality instead gives $|\chi(c+iqz)|^2 \geq q^{2q}(3/4)^q$, and $|f(c+iqz)|^2 \geq q^{2l}$. On $|z| \leq \epsilon$, LET4 gives the respective upper bounds $q^{2q}(2\epsilon^2)^q$ and $q^{2l}(2\epsilon^2)^l$. The coefficient polynomial is exactly the constant one. In particular its scalar value is bounded by $8^{2q} \int_1^2 1 dz$, with no missing coefficient directions. Integrating over the inner segment and dividing by the positive bulk subsegment gives for both actual weights

$$H^I/H^B \leq \kappa_{I,1} := \frac{2\epsilon}{\sqrt{\cos 1}} \left(\frac{4}{3} 2^{-13} e^{2\pi} \right)^q. \quad (\text{LET12})$$

Indeed the numerator/denominator ratio before enlarging it is $2\epsilon(\cos 1)^{-1/2}(4/3)^q e^{2\pi q} (2\epsilon^2)^{q+s}$, $s = 0, l$. Multiplication by $8^{2q} \geq 1$ gives LET12 because $2\epsilon^2 8^2 = 2^{-13}$ and $(2\epsilon^2)^s \leq 1$. For the reference powers the corresponding stronger constant is $2\epsilon(\cos 1)^{-1/2} \epsilon^{2s} (2^{-14} e^{2\pi})^q$, hence is at most LET12.

For the far estimate retain $D_s = 2q + 2s \leq 5q$. At $z \geq 64$, integration of $\log(z/64) \leq (z-64)/64$ gives

$$\int_{64}^{\infty} z^{D_s} e^{-qz} dz \leq \frac{64^{D_s} e^{-64q}}{q - D_s/64}, \quad q - D_s/64 \geq 59q/64. \quad (\text{LET13})$$

For the actual weights, LET7 on the bulk subsegment gives the lower factor $e^{-\eta}$, where $\eta = -a_k + \delta_\chi(1)$, and $\delta_\chi(x) = q\tau_x^2/[2(1-\tau_x)]$. The far upper weights have the extra factors $e^{\delta_\chi(64)+\Delta_{f,s}(64)}$, where $\Delta_{f,0} = 0$, $\Delta_{f,l}(x) = S_2/(q^2x^2)$. Using LET11, both real tails, and the Jacobian $dy = q dz$, their ratio is at most

$$\frac{2}{\sqrt{\cos 1}} \frac{e^{2\pi q + \eta + \delta_\chi(64) + \Delta_{f,s}(64)} 64^{D_s} e^{-64q}}{q - D_s/64}. \quad (\text{LET14})$$

No polynomial-growth factor is needed for the constant coefficient. The original domain implies $\tau_1 \leq 1/4$ and $k \geq 5$. It gives separately

$$-a_k \leq 7q/60, \quad \delta_\chi(1) \leq q/24, \quad \delta_\chi(64) \leq q/24, \quad \Delta_{f,l}(64) \leq q/8192.$$

The last follows from $S_2 \leq 4l^3$ and $2l \leq q$. Their sum is less than $q/4$. Using $64 - 2\pi - 5 \log 64 > 20$, LET13–14 therefore give

$$H^F/H^B \leq \frac{128}{59q\sqrt{\cos 1}} e^{-19q} \leq \kappa_F := \frac{512q}{59\sqrt{\cos 1}} e^{-19q}. \quad (\text{LET15})$$

The elementary displayed strict exponent bound follows from $\pi < 4$ and $\log 2 < 1$. For the reference powers all three correction factors in LET14 equal one, so the same inequality follows directly. Thus the four actual scalar corrections obey

$$0 \leq t_H \leq L_1 := \log(1 + \kappa_{I,1} + \kappa_F). \quad (\text{LET16})$$

The constant $\kappa_{I,1}$ decays exponentially: as proved by the positive series for $\log 2$, $\log 2 > 2/3$; also $\log(4/3) < 1/3$ and $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2) dx > 0$. Their substitution gives $\log((4/3)2^{-13}e^{2\pi}) < -43/21 < -2$.

The complete exact low-endpoint transfer is now

$$\log \mathfrak{r}_\chi - \log \frac{m_{2q+2l}}{m_{2q}} = d_\chi + d_f + t_{H_{\chi,f}} - t_{H_{\chi,0}} - t_{H_{\text{pow},l}} + t_{H_{\text{pow},0}}. \quad (\text{LET17})$$

It follows by writing each of the four full integrals as its own bulk integral times e^{t_H} . In particular, with $a_0 = E_0 + 2L_1$ and $b_0 = a_0 + \Delta_f$,

$$\boxed{-a_0 \leq \log \mathfrak{r}_\chi - \log \frac{m_{2q+2l}}{m_{2q}} \leq b_0.} \quad (\text{LET18})$$

The exact one-sided numerator correction occurs only in the upper bound. Since

$$|a_k| \leq |d - g|/(3e), \quad \delta_\chi \leq \frac{2(d+g)^2}{3e^3k^2\epsilon^4}, \quad \Delta_f \leq \frac{1}{16e^2k\epsilon^2},$$

the finite errors in LET18 are $O(1)$ at fixed $m = 1, \delta, \gamma$ and $O(k^{-1})$ for each fixed $m \geq 2, \delta, \gamma$. The last displayed bound uses $S_2 \leq k^3/16$ and $q \geq ek^2$. Both tail corrections have the already proved exponential decay.

4 The whole original signed Gamma functional

Retain the full high reference matrices

$$(\mathbf{M}_{a,r})_{ij} = m_{2a+i+j}, \quad \mathcal{H}_{q,l,r} = \det \mathbf{M}_{q+l,r} / \det \mathbf{M}_{q,r}, \quad r = q, q+1.$$

Let

$$P_k = \log D_{l,q-1} + \log D_{l,q} - \log D_{l,2q-1} - \log D_{l,2q}, \quad D_{l,N} = \prod_{a=0}^N \prod_{j=0}^{2l-1} (a+j+1/2),$$

$$W_k = P_k - \log \frac{m_{2q+2l}}{m_{2q}} + \log \mathcal{H}_{q,l,q} + \log \mathcal{H}_{q,l,q+1}. \quad (\text{LET19})$$

All factors of the original density are retained in the moment matrices; CTR53 proves their exact rational moment calculation with mass factors displayed before cancellation. The exact old identity is $\mathfrak{T}_k = P_k - \log \mathfrak{r}_\chi + \log \mathfrak{R}_{2q-1} + \log \mathfrak{R}_{2q}$.

Write the proved high-endpoint constants from CTR45–48 as

$$A = 2E_{\chi,l} + 2(L_q + L_{q+1}), \quad B = A + (2q + 1)\Delta_f, \quad E_{\chi,l} = lE_0.$$

Their complete signed sum has error in $[-A, B]$. The minus sign of the low endpoint reverses LET18. Therefore

$$\boxed{W_k - A - b_0 \leq \mathfrak{T}_k \leq W_k + B + a_0}, \quad \boxed{-W_k - B - a_0 \leq R_k^0 - R_k^\sigma \leq -W_k + A + b_0}. \quad (\text{LET20})$$

This is a direct substitution of exact determinant identities and finite proved errors. In particular the entire Gamma reference W_k depends only on the original integers q, l and the same fixed Gamma density. The original δ, γ remain in its error constants, finite threshold, original quotient, and all exact relation maps. Both sides have error $O(k) = o(q)$; at every fixed $m \geq 2$ the error is $O(1)$. No value is assigned here to the remaining explicit W_k .

Finally, preserve the exact earlier arithmetic notation $\Delta_\Gamma = \delta_\sigma - \mathfrak{T}_k$ and $B_{\text{ar}} = B_0 + \Delta_\Gamma$, with $-a_- \leq \delta_\sigma \leq a_+$. Adding these signed intervals gives the completed receiving formula

$$B_0 - W_k - B - a_0 - a_- \leq B_{\text{ar}} \leq B_0 - W_k + A + b_0 + a_+. \quad (\text{LET21})$$

The baseline B_0 in LET21 is the existing arithmetic quantity, whereas the lowercase a_0, b_0 are the low-source errors explicitly defined above. Subtracting the original $4q \log(D_h k)$ gives the corresponding interval for the unchanged HC residual.

5 Original moments and the two sheets of the squared coordinate

Retain $k = 4l + 1$, $l \geq 1$, $m \geq 1$, $e = 1 + k(m - 1)$, $q = e(k + 1)^2$, $c = k/2$, and the original measure and its mass

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad c_\sigma = \int_{\mathbb{R}} d\sigma = \sqrt{2\pi}. \quad (\text{PHT1})$$

Use distinct names for the ordinary and even moments:

$$m_j = \int_{\mathbb{R}} y^j d\sigma(y), \quad \mu_j = m_{2j}, \quad M_{a,r} = (m_{2a+i+j})_{0 \leq i,j < r}, \quad H_n^{(a)} = \det(\mu_{a+i+j})_{0 \leq i,j < n}. \quad (\text{PHT2})$$

Here a, n are nonnegative integers, $r \geq 1$, and $H_0^{(a)} = 1$. All moments are finite: rotation of Euler's Gamma integral through any angle $0 < \theta < \pi/2$ gives

$$\sigma(y) \leq \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos \theta}} e^{-\theta|y|}.$$

For the rotation use the argument $\text{sgn}(y)\theta$; the factor contributed by the power is $e^{-\theta|y|/2}$, and the absolute remaining Euler integral is $\Gamma(1/4)(\cos \theta)^{-1/4}$ before squaring. The small and large circular arcs vanish by the positive real exponent and by $\cos \theta > 0$. This proves the displayed bound and its constants directly.

The positive-axis and negative-axis maps to $t = y^2$ have inverse branches $y = \sqrt{t}$ and $y = -\sqrt{t}$. Each branch pushes the original measure to

$$d\nu_+(t) = d\nu_-(t) = \frac{\sigma(\sqrt{t})}{2\sqrt{t}} dt, \quad t > 0.$$

The reversed orientation of the negative branch changes the integration limits as well as its Jacobian, so its measure is positive with the same factor $1/2$. Thus the full pushforward is

$$d\nu(t) = d\nu_+(t) + d\nu_-(t) = \frac{\sigma(\sqrt{t})}{\sqrt{t}} dt, \quad \int_0^\infty d\nu = c_\sigma, \quad \mu_j = \int_0^\infty t^j d\nu(t). \quad (\text{PHT3})$$

No half-axis contribution has been treated as the whole mass.

Every polynomial has a unique decomposition $P(y) = P_e(y^2) + yP_o(y^2)$. The exact coefficient map $(P_e, P_o) \mapsto P_e(y^2) + yP_o(y^2)$ has inverse

$$P_e(t) = \frac{P(\sqrt{t}) + P(-\sqrt{t})}{2}, \quad P_o(t) = \frac{P(\sqrt{t}) - P(-\sqrt{t})}{2\sqrt{t}}. \quad (\text{PHT4})$$

Adding the two branch integrals cancels their cross terms and gives

$$\int_{\mathbb{R}} |P(y)|^2 y^{2a} d\sigma(y) = \int_0^\infty |P_e(t)|^2 t^a d\nu(t) + \int_0^\infty |P_o(t)|^2 t^{a+1} d\nu(t). \quad (\text{PHT5})$$

These same formulas extend to a unitary map of the corresponding L^2 spaces: the inverse in PHT4 is defined for almost every $t > 0$, and PHT5 proves both its range and norm. The point $t = 0$ has zero measure. On the exact multiplication domains this map transports the original real coordinate operator to

$$M_y \longleftrightarrow \begin{pmatrix} 0 & M_t \\ I & 0 \end{pmatrix}, \quad M_y^2 \longleftrightarrow \begin{pmatrix} M_t & 0 \\ 0 & M_t \end{pmatrix}. \quad (\text{PHT6})$$

Indeed multiplication of $P_e(y^2) + yP_o(y^2)$ by y gives $(tP_o)(y^2) + yP_e(y^2)$. Thus the two parity blocks retain their exact coupling under the original coordinate. Explicitly, in the base Hilbert space

$$\mathcal{K}_a = L^2(t^a d\nu) \oplus L^2(t^{a+1} d\nu),$$

the first operator has domain

$$\{(u, v) \in \mathcal{K}_a : u \in L^2(t^{a+1}d\nu), \quad v \in L^2(t^{a+2}d\nu)\}$$

and acts by $(u, v) \mapsto (tv, u)$. Its squared operator has domain

$$\{(u, v) \in \mathcal{K}_a : u \in L^2(t^{a+2}d\nu), \quad v \in L^2(t^{a+3}d\nu)\}$$

and acts by $(u, v) \mapsto (tu, tv)$. These are exactly the conditions that the displayed images belong to $\mathcal{K}_a$. For the squared domain the intermediate domain conditions follow from $t \leq 1 + t^2$. PHT5 then identifies these domains with $\{P : yP \in L^2(y^{2a}d\sigma)\}$ and $\{P : y^2P \in L^2(y^{2a}d\sigma)\}$, respectively, including the membership of P in that base space.

6 The complete finite parity congruence

Set $n_e = \lceil r/2 \rceil, n_o = \lfloor r/2 \rfloor$. Let Π_r have columns $e_0, e_2, \dots, e_{2n_e-2}$ followed by $e_1, e_3, \dots, e_{2n_o-1}$ in the original coefficient frame. Then $\Pi_r^{-1} = \Pi_r^\top$ and $\det \Pi_r = (-1)^{n_e(n_e-1)/2}$: the j -th even index crosses exactly j smaller odd indices. Evenness of σ makes each odd moment zero. Entrywise substitution therefore proves

$$\Pi_r^\top M_{a,r} \Pi_r = \begin{pmatrix} (\mu_{a+i+j})_{i,j < n_e} & 0 \\ 0 & (\mu_{a+1+i+j})_{i,j < n_o} \end{pmatrix}. \quad (\text{PHT7})$$

This is a congruence in the original coefficient space. Its two determinant signs cancel, so with $D_{a,r} = \det M_{a,r}$,

$$D_{a,2n} = H_n^{(a)} H_n^{(a+1)}, \quad D_{a,2n+1} = H_{n+1}^{(a)} H_n^{(a+1)}. \quad (\text{PHT8})$$

Every factor is strictly positive: its quadratic form is the integral of a nonzero polynomial's squared modulus against a positive density on $(0, \infty)$.

The original S frame is also retained. The map $P(S) \mapsto P(c + iy)$ has coefficient matrix

$$E_{j,n} = \binom{n}{j} c^{n-j} (\sqrt{-1})^j \quad (j \leq n), \quad E^{-1} : Q(y) \mapsto Q((S - c)/i), \quad \det E = i^{r(r-1)/2}. \quad (\text{PHT9})$$

Entries below the displayed triangular range are zero. The source Gram in the S frame is exactly $E^* M_{a,r} E$, and its determinant is $D_{a,r}$ because $|\det E| = 1$. Multiplication by $(S - c)^a$ contributes the exact scalar i^a under this map; its squared modulus is one. These identities identify the moments in PHT2 with the actual monomial Gamma relation subspaces, including their degree.

7 The two original high degrees reduce to one positive shift

The stipulated $q = e(k+1)^2$ is even; put $n = q/2$. Define

$$Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2, \quad A_n^{(a)} = H_n^{(a+1)} / H_n^{(a)} > 0. \quad (\text{PHT10})$$

At the two required ranks $r = q, q+1$, PHT8 proves the exact combination

$$\frac{D_{q+l,q}}{D_{q,q}} \frac{D_{q+l,q+1}}{D_{q,q+1}} = \frac{Z_{q+l}}{Z_q}, \quad \frac{Z_{q+1}}{Z_a} = A_n^{(a)} A_{n+1}^{(a)} (A_n^{(a+1)})^2. \quad (\text{PHT11})$$

Hence its logarithm is exactly the finite sum

$$\log \frac{Z_{q+l}}{Z_q} = \sum_{j=0}^{l-1} \{ \log A_n^{(q+j)} + \log A_{n+1}^{(q+j)} + 2 \log A_n^{(q+j+1)} \}. \quad (\text{PHT12})$$

This is a contiguous-shift expression for the original two high determinants, with their unchanged ranks. The original mass factor in each $H_n^{(a)}$ is c_σ^n , so Z_a contains $c_\sigma^{4n+1} = c_\sigma^{2q+1}$. It cancels only after the numerator and denominator in PHT11 are formed.

8 Original Toda time and the squared-coordinate time

For $u \geq 0$, define the actual pushed-forward family

$$d\nu_{a,u}(t) = t^a e^{-ut} d\nu(t), \quad H_n^{(a)}(u) = \det \left(\int_0^\infty t^{a+i+j} e^{-ut} d\nu(t) \right)_{i,j < n}. \quad (\text{PHT13})$$

This is precisely the original y -source multiplied by e^{-uy^2} , pushed forward through its two branches. All moment derivatives at $u = 0$ exist from the right by dominated convergence using the original higher moments. For $\text{Re } u > 0$ they are holomorphic, by the same domination on compact subsets. The real family with $u < 0$ does not have finite mass: Gamma reflection and the Euler bound give $\sigma(y) \geq C e^{-\pi|y|}$ with $C = \pi/\Gamma(3/4)^2 > 0$, and $e^{|u|t - \pi\sqrt{t}}$ grows exponentially for sufficiently large t . This proves the actual time domain rather than assigning an unjustified two-sided strip to the quadratic flow. The reflection formula here follows on $0 < \text{Re } z < 1$ by changing $x = tv$ in the Euler double integral, obtaining $\Gamma(z)\Gamma(1-z) = \int_0^\infty t^{z-1}/(1+t) dt$. A keyhole contour with argument in $(0, 2\pi)$ has residue $e^{i\pi(z-1)}$ at -1 and straight-segment factor $1 - e^{2\pi iz}$; its vanishing circular arcs give the value $\pi/\sin(\pi z)$. At $z = 1/4 + iy/2$, take squared moduli and use $|\Gamma(3/4 - iy/2)| \leq \Gamma(3/4)$ from Euler's integral. Since $|\sin(\pi/4 + i\pi y/2)|^2 = \cosh(\pi y)/2$, this proves the stated lower bound.

Let $p_j^{(a,u)}$ be the monic orthogonal polynomials in this measure, with norms

$$h_j = H_{j+1}^{(a)}(u)/H_j^{(a)}(u).$$

Their coefficients are obtained by inverting the positive finite moment Gram, so the moment differentiability just proved also justifies each coefficient derivative below. Expansion of tp_j and orthogonality give

$$tp_j = p_{j+1} + b_j p_j + \alpha_j p_{j-1}, \quad \alpha_j = h_j/h_{j-1} > 0 \ (j \geq 1), \quad \alpha_0 = 0. \quad (\text{PHT14})$$

All other lower coefficients vanish because $\langle tp_j, p_i \rangle = \langle p_j, tp_i \rangle = 0$ for $i < j-1$; comparison against p_{j-1} gives α_j . Differentiating orthogonality under the negative tilt proves $\partial_u p_j = \alpha_j p_{j-1}$, since its coefficient against every p_i is that of tp_j for $i < j$. Also $\partial_u h_j = -b_j h_j$. Differentiating $b_j = h_j^{-1} \int tp_j^2 d\nu_{a,u}$ and using PHT14 gives

$$\partial_u b_j = \alpha_j - \alpha_{j+1}, \quad \partial_u \alpha_j = \alpha_j (b_{j-1} - b_j). \quad (\text{PHT15})$$

In detail the derivative of the numerator of b_j is $2\alpha_j h_j - \|tp_j\|^2 = h_j(\alpha_j - \alpha_{j+1} - b_j^2)$, and differentiating h_j^{-1} supplies the missing b_j^2 . Since $H_n = \prod_{j < n} h_j$, telescoping PHT15 proves

$$(\log H_n^{(a)}(u))'' = \frac{H_{n+1}^{(a)}(u)H_{n-1}^{(a)}(u)}{H_n^{(a)}(u)^2} > 0 \quad (n \geq 1), \quad (\text{PHT16})$$

or equivalently $H_n H_n'' - (H_n')^2 = H_{n+1} H_{n-1}$. At $u = 0$ these derivatives are the proved right derivatives. Thus the coefficient and sign of the Hankel-Toda equation are exactly those of the existing original-source Toda calculation.

There is an explicit map to that calculation's original linear tilt as well. For real θ , on $L^2(y^{2a} d\sigma)$, multiplication by $e^{\theta y/2}$ is the isometry from the source weighted by $e^{\theta y}$ to the original source, with inverse given by the negative exponential. Under PHT4–5 the same map is the following exact operator matrix on the two branches:

$$\begin{pmatrix} \cosh(\theta\sqrt{t}/2) & \sqrt{t} \sinh(\theta\sqrt{t}/2) \\ t^{-1/2} \sinh(\theta\sqrt{t}/2) & \cosh(\theta\sqrt{t}/2) \end{pmatrix}. \quad (\text{PHT17})$$

Substituting $y = \sqrt{t}$ and $y = -\sqrt{t}$ proves each entry and its inverse at $-\theta$. To specify the weighted source and its full cross Gram, write this matrix as $B_{\theta/2}(t)$. The two branch integrals give the pulled-back form

$$\int_0^\infty \begin{pmatrix} \bar{u} & \bar{v} \end{pmatrix} G_\theta(t) \begin{pmatrix} u \\ v \end{pmatrix} t^a d\nu(t), \quad G_\theta(t) = \begin{pmatrix} \cosh(\theta\sqrt{t}) & \sqrt{t} \sinh(\theta\sqrt{t}) \\ \sqrt{t} \sinh(\theta\sqrt{t}) & t \cosh(\theta\sqrt{t}) \end{pmatrix}. \quad (\text{PHT17a})$$

For real θ , direct matrix multiplication, using $\cosh^2 x + \sinh^2 x = \cosh(2x)$ and $2 \sinh x \cosh x = \sinh(2x)$, proves

$$B_{\theta/2}(t)^* \text{diag}(1, t) B_{\theta/2}(t) = G_\theta(t), \quad B_{-\theta/2}(t) B_{\theta/2}(t) = I.$$

Thus the space of measurable pairs with finite PHT17a norm is mapped isometrically and onto $\mathcal{K}_a$ by $B_{\theta/2}$, with inverse $B_{-\theta/2}$; the equality of forms proves the domain and range even when these matrices are unbounded on the base space. If viewed as an operator on $\mathcal{K}_a$ itself, its domain consists precisely of the pairs in $\mathcal{K}_a$ whose PHT17a integral is finite. The branch map in PHT4 identifies that source with the full original space $L^2(e^{\theta y} y^{2a} d\sigma)$. The quadratic tilt in PHT13 instead gives the diagonal multiplication $e^{-ut/2}$ in both components. Both are the specified functions of the operator in PHT6. This proves the relation between the two times on the original spaces and keeps the parity mixing of the original linear time.

For real $|\theta| < \pi/2$ the tilted y -moments and their derivatives converge: in the earlier Gamma bound choose $|\theta| < \theta_0 < \pi/2$. The same orthogonal-polynomial derivative calculation, with real variable y and positive tilt $e^{\theta y}$, applies to the exactly defined matrix $M_{a,r}(\theta)_{ij} = \int_{\mathbb{R}} y^{2a+i+j} e^{\theta y} d\sigma(y)$ and proves

$$(\log \det M_{a,r}(\theta))''|_{\theta=0} = h_r^{y,a} / h_{r-1}^{y,a}. \quad (\text{PHT18})$$

Here the superscript y, a denotes the full original real-line monic norms. Parity and PHT5 identify them exactly as $h_{2j}^{y,a} = H_{j+1}^{(a)} / H_j^{(a)}$ and $h_{2j+1}^{y,a} = H_{j+1}^{(a+1)} / H_j^{(a+1)}$. Consequently the two original Toda curvatures have the explicit positive expressions

$$\begin{aligned} \frac{h_{2n}^{y,a}}{h_{2n-1}^{y,a}} &= \frac{H_{n+1}^{(a)} H_{n-1}^{(a+1)}}{H_n^{(a)} H_n^{(a+1)}}, \\ \frac{h_{2n+1}^{y,a}}{h_{2n}^{y,a}} &= \frac{H_{n+1}^{(a+1)} H_n^{(a)}}{H_n^{(a+1)} H_{n+1}^{(a)}}. \end{aligned} \quad (\text{PHT19})$$

These are exact specializations of the original norm-ratio Toda controls at the ranks needed by the programme. The contiguous identities and their positive computation follow in the attached complete derivation.

9 Exact contiguous Gamma determinants and positive Christoffel evolution

HCT1. Original measure and the typed square map. Throughout this section

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \gamma_0 = \sqrt{2\pi}, \quad y \in \mathbb{R}.$$

Let $s : \mathbb{R} \rightarrow [0, \infty)$ be $s(y) = y^2$ and let $\nu = s_* \sigma$. For every nonnegative Borel function g , change of variable on each half of the original real line gives

$$\int_{\mathbb{R}} g(y^2) d\sigma(y) = \int_0^\infty g(t) \frac{|\Gamma(1/4 + i\sqrt{t}/2)|^2}{2\pi\sqrt{t}} dt. \quad (\text{HCT1})$$

Thus the factor $t^{-1/2}$ includes both preimages of $t > 0$. For integers $a, n \geq 0$ put

$$d\nu_a(t) = t^a d\nu(t), \quad \mu_n = \int_{\mathbb{R}} y^{2n} d\sigma(y) = \int_0^\infty t^n d\nu(t), \quad H_n^{(a)} = \det[\mu_{a+i+j}]_{i,j=0}^{n-1}, \quad H_0^{(a)} = 1.$$

No factor γ_0 is removed from any $H_n^{(a)}$.

All moments exist. Indeed, rotation by $\theta \operatorname{sgn}(y)$ in Euler's Gamma integral, with $0 < \theta < \pi/2$, gives

$$|\Gamma(1/4 + iy/2)| \leq \Gamma(1/4) (\cos \theta)^{-1/4} e^{-\theta|y|/2}.$$

The rotation is justified on a truncated sector: the small circular arc is $O(\epsilon^{1/4})$, and the large arc tends to zero because $\cos \theta > 0$. Squaring proves exponential decay of σ . The Gamma function is continuous and nonzero at $1/4$, so the density is positive on an interval about zero. Every nonzero polynomial has positive squared integral against every ν_a ; consequently

$$H_n^{(a)} > 0 \quad (a \geq 0, n \geq 0). \quad (\text{HCT2})$$

HCT2. Gamma mass, moments, and the full real-line polynomial norms. The constants needed below can be derived directly from the original Gamma weight. For $\lambda = 1/4$ set $g(x) = e^{\lambda x - e^x}$. With the Fourier convention $\widehat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{i\xi x} dx$, Euler's integral gives $\widehat{g}(\xi) = \Gamma(\lambda + i\xi)$. The function g and all its derivatives decay faster than every inverse power at both ends. Fourier inversion of its autocorrelation therefore gives

$$\int_{\mathbb{R}} e^{iu\xi} |\Gamma(\lambda + i\xi)|^2 d\xi = 2\pi \int_{\mathbb{R}} g(x + u/2) g(x - u/2) dx = \frac{2\pi \Gamma(2\lambda)}{(2 \cosh(u/2))^{2\lambda}}.$$

The last equality follows by $v = e^x$ in the integral on the right. Substituting $y = 2\xi$ and $u = 2w$ gives the exact characteristic function

$$\int_{\mathbb{R}} e^{iwy} d\sigma(y) = \gamma_0(\cosh w)^{-1/2} \quad (w \in \mathbb{R}).$$

The exponential bound in HCT1 permits its analytic continuation to the open strip $|\operatorname{Im} w| < \pi/2$. In particular

$$\int_{\mathbb{R}} e^{zy} d\sigma(y) = \gamma_0(\cos z)^{-1/2} \quad (|z| < \pi/2), \quad \mu_n = \gamma_0(2n)! [z^{2n}] (\cos z)^{-1/2}. \quad (\text{HCT3})$$

The square root in (HCT3) is positive at $z = 0$.

Define monic polynomials $p_j(y)$ by the convergent generating function on $|z| < 1$, with the branches analytic at zero and equal to their real values there,

$$\sum_{j=0}^{\infty} p_j(y) \frac{z^j}{j!} = (1 + z^2)^{-1/4} \exp(y \arctan z). \quad (\text{HCT4})$$

For sufficiently small real z, w , (HCT3) and

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

show that the integral of the product of the two generating functions is $\gamma_0(1 - zw)^{-1/2}$. Derivatives may pass under the integral by the exponential bound in HCT1. Comparison of coefficients proves

$$\int_{\mathbb{R}} p_i(y) p_j(y) d\sigma(y) = \mathbf{1}_{i=j} \gamma_0 j! (1/2)_j. \quad (\text{HCT5})$$

Here $(b)_j = b(b+1) \cdots (b+j-1)$ and $(b)_0 = 1$. Differentiating (HCT4) gives, with $p_{-1} = 0$,

$$y p_j(y) = p_{j+1}(y) + j(j-1/2) p_{j-1}(y), \quad p_0 = 1. \quad (\text{HCT6})$$

In particular p_{2n} is even and p_{2n+1} is odd. This also follows directly from $p_j(-y) = (-1)^j p_j(y)$ in (HCT4).

HCT3. Monic polynomials on the actual square-pushforward. Let $P_n^{(a)}(t)$ be the monic polynomial orthogonal to $1, t, \dots, t^{n-1}$ in $L^2(\nu_a)$, and write $h_n^{(a)} = \int (P_n^{(a)})^2 d\nu_a$. Positivity (HCT2) gives existence and uniqueness by finite Gram elimination. The determinant formula and its constant term are

$$P_n^{(a)}(t) = \frac{1}{H_n^{(a)}} \det \begin{bmatrix} \mu_a & \mu_{a+1} & \cdots & \mu_{a+n} \\ \mu_{a+1} & \mu_{a+2} & \cdots & \mu_{a+n+1} \\ \vdots & \vdots & & \vdots \\ \mu_{a+n-1} & \mu_{a+n} & \cdots & \mu_{a+2n-1} \\ 1 & t & \cdots & t^n \end{bmatrix}, \quad P_n^{(a)}(0) = (-1)^n \frac{H_n^{(a+1)}}{H_n^{(a)}}. \quad (\text{HCT7})$$

For $n = 0$ this means $P_0^{(a)} = 1$. The last-row expansion in the first column proves the displayed sign $(-1)^n$. Orthogonality follows by integrating against t^j , which duplicates a row for $0 \leq j < n$. A monic triangular change from monomials to the $P_j^{(a)}$ has determinant one, so

$$H_n^{(a)} = \prod_{j=0}^{n-1} h_j^{(a)}, \quad h_n^{(a)} = \frac{H_{n+1}^{(a)}}{H_n^{(a)}} > 0. \quad (\text{HCT8})$$

Define

$$c_n^{(a)} = \frac{H_{n+1}^{(a+1)} H_n^{(a)}}{H_{n+1}^{(a)} H_n^{(a+1)}} > 0 \quad (n \geq 0), \quad (\text{HCT9})$$

$$d_n^{(a)} = \frac{H_{n+1}^{(a)} H_{n-1}^{(a+1)}}{H_n^{(a)} H_n^{(a+1)}} > 0 \quad (n \geq 1), \quad d_0^{(a)} = 0. \quad (\text{HCT10})$$

The exact Christoffel and inverse connection identities are

$$tP_n^{(a+1)}(t) = P_{n+1}^{(a)}(t) + c_n^{(a)} P_n^{(a)}(t), \quad (\text{HCT11})$$

$$P_n^{(a)}(t) = P_n^{(a+1)}(t) + d_n^{(a)} P_{n-1}^{(a+1)}(t) \quad (n \geq 1). \quad (\text{HCT12})$$

For $n = 0$ the second identity reads $1 = 1$. To prove (HCT11), (HCT7) gives $P_{n+1}^{(a)}(0) + c_n^{(a)} P_n^{(a)}(0) = 0$. Division by t is consequently polynomial division with zero remainder. The quotient is monic of degree n and has zero ν_{a+1} inner product with t^j for $j < n$; uniqueness proves the identity. Taking its ν_a inner product with $P_n^{(a+1)}$ proves

$$h_n^{(a+1)} = c_n^{(a)} h_n^{(a)}. \quad (\text{HCT13})$$

Indeed the $P_{n+1}^{(a)}$ term has zero inner product, while $P_n^{(a+1)} - P_n^{(a)}$ has degree at most $n - 1$. For (HCT12), $P_n^{(a)} - P_n^{(a+1)}$ has degree at most $n - 1$. Its ν_{a+1} inner products with all polynomials of degree at most $n - 2$ vanish, because multiplication of such a polynomial by t has degree at most $n - 1$. It is therefore a multiple of $P_{n-1}^{(a+1)}$. The numerator of that coefficient is

$$\int P_n^{(a)} P_{n-1}^{(a+1)} d\nu_{a+1} = \int P_n^{(a)} t P_{n-1}^{(a+1)} d\nu_a = h_n^{(a)},$$

using (HCT11) at $n - 1$. Dividing by $h_{n-1}^{(a+1)}$ gives exactly (HCT10), including its positive sign.

HCT4. Positive discrete Toda identities. Combining (HCT11) and (HCT12) twice, entirely within the original polynomial spaces, gives

$$tP_n^{(a)} = P_{n+1}^{(a)} + (c_n^{(a)} + d_n^{(a)}) P_n^{(a)} + d_n^{(a)} c_{n-1}^{(a)} P_{n-1}^{(a)}. \quad (\text{HCT14})$$

At $n = 0$ the last term is absent. Expanding instead in the $a + 1$ basis gives

$$tP_n^{(a+1)} = P_{n+1}^{(a+1)} + (c_n^{(a)} + d_{n+1}^{(a)})P_n^{(a+1)} + c_n^{(a)}d_n^{(a)}P_{n-1}^{(a+1)}.$$

Coefficient comparison with (HCT14) at $a + 1$ proves

$$c_n^{(a+1)} + d_n^{(a+1)} = c_n^{(a)} + d_{n+1}^{(a)} \quad (n \geq 0), \quad (\text{HCT15})$$

$$c_{n-1}^{(a+1)}d_n^{(a+1)} = c_n^{(a)}d_n^{(a)} \quad (n \geq 1). \quad (\text{HCT16})$$

All nonzero factors in these identities are positive. In the notation $b_n^{(a)} = c_n^{(a)} + d_n^{(a)}$ and $u_n^{(a)} = d_n^{(a)}c_{n-1}^{(a)}$, the two recurrences give the complete b, u coefficients and their shifted values, including $b_n^{(a+1)} = c_n^{(a)} + d_{n+1}^{(a)}$ and $u_n^{(a+1)} = c_n^{(a)}d_n^{(a)}$.

Here is the exact polynomial meaning of the matrix factorization. Use column vectors

$$\mathbf{P}^{(a)} = (P_0^{(a)}, P_1^{(a)}, \dots)^T$$

as formal polynomial sequences. Let L_a have diagonal 1 and subdiagonal $(L_a)_{n,n-1} = d_n^{(a)}$, and let U_a have diagonal $c_n^{(a)}$ and superdiagonal 1. Every row has finite support. Then (HCT11)–(HCT12) state

$$\mathbf{P}^{(a)} = L_a \mathbf{P}^{(a+1)}, \quad t\mathbf{P}^{(a+1)} = U_a \mathbf{P}^{(a)}, \quad J_a = L_a U_a, \quad J_{a+1} = U_a L_a. \quad (\text{HCT17})$$

Consequently $J_{a+1}U_a = U_a J_a$ and $J_a L_a = L_a J_{a+1}$. These are row-finite polynomial identities; no infinite-dimensional bounded-operator assertion or truncation of an endpoint row is needed. At a finite degree the identities retain every displayed P_{n+1} term. The scalar multiplication map is literally $t : \mathbb{R}[t]_{\leq n} \rightarrow \mathbb{R}[t]_{\leq n+1}$.

HCT5. Exact initial Gamma data and a subtraction-free finite evolution. Parity, (HCT1), and uniqueness give the exact source identifications

$$p_{2n}(y) = P_n^{(0)}(y^2), \quad p_{2n+1}(y) = yP_n^{(1)}(y^2). \quad (\text{HCT18})$$

Thus (HCT5) gives, with the original mass retained,

$$h_n^{(0)} = \gamma_0(2n)!(1/2)_{2n}, \quad H_n^{(0)} = \gamma_0^n \prod_{j=0}^{n-1} (2j)!(1/2)_{2j}, \quad (\text{HCT19})$$

$$h_n^{(1)} = \gamma_0(2n+1)!(1/2)_{2n+1}, \quad H_n^{(1)} = \gamma_0^n \prod_{j=0}^{n-1} (2j+1)!(1/2)_{2j+1}. \quad (\text{HCT20})$$

Equations (HCT9)–(HCT10) therefore yield the exact positive rational starting values

$$c_n^{(0)} = (2n+1)(2n+1/2), \quad d_n^{(0)} = 2n(2n-1/2), \quad n \geq 0. \quad (\text{HCT21})$$

For a fixed step $a \rightarrow a + 1$ introduce auxiliary numbers e_n by

$$e_0 = c_0^{(a)}, \quad d_0^{(a+1)} = 0, \quad c_0^{(a+1)} = e_0 + d_1^{(a)}, \quad (\text{HCT22})$$

$$e_n = \frac{c_n^{(a)}e_{n-1}}{e_{n-1} + d_n^{(a)}}, \quad d_n^{(a+1)} = \frac{c_n^{(a)}d_n^{(a)}}{e_{n-1} + d_n^{(a)}}, \quad c_n^{(a+1)} = e_n + d_{n+1}^{(a)} \quad (n \geq 1). \quad (\text{HCT23})$$

Every denominator and every $e_n, c_n^{(a+1)}$ is strictly positive; $d_n^{(a+1)} > 0$ for $n \geq 1$. These formulas use only positive sums, products, and quotients. To verify them, (HCT15) at zero gives (HCT22). If $c_{n-1}^{(a+1)} = e_{n-1} + d_n^{(a)}$, then (HCT16) gives the middle formula in (HCT23). The first formula implies

$e_n + d_n^{(a+1)} = c_n^{(a)}$; substituting this in (HCT15) gives the last formula. This proves the recurrence and the claimed positivity by induction.

It follows, without determinant cancellation, that for any finite integer $a, n \geq 0$

$$H_n^{(a)} = H_n^{(0)} \prod_{s=0}^{a-1} \prod_{j=0}^{n-1} c_j^{(s)}. \quad (\text{HCT24})$$

Indeed $H_n^{(a+1)}/H_n^{(a)} = \prod_{j < n} c_j^{(a)}$ follows from (HCT8) and (HCT13), and then telescopes in a . An initial array through index $N + A$ suffices to compute all $c_n^{(a)}, d_n^{(a)}$ with $a \leq A, n \leq N$: one may retain indices $0 \leq n \leq N + A - a$ after step a and shrink the retained interval by one at each subsequent step. Thus the formulas give a finite exact calculation for arbitrarily specified original ranks and shifts. The assertion here concerns an exact finite recurrence, with no unproved assertion about its computational cost or large-rank limit.

HCT6. Sign-correct contiguous determinants. For $n \geq 1$, apply the Desnanot–Jacobi identity to the $(n+1) \times (n+1)$ matrix $[\mu_{a+i+j}]_{i,j=0}^n$, deleting its first and last rows and columns. The two principal minors are $H_n^{(a+2)}$ and $H_n^{(a)}$, the two cross minors are both $H_n^{(a+1)}$, and the central minor is $H_{n-1}^{(a+2)}$. Their original index order is retained. Hence

$$H_n^{(a)} H_n^{(a+2)} - (H_n^{(a+1)})^2 = H_{n+1}^{(a)} H_{n-1}^{(a+2)} > 0. \quad (\text{HCT25})$$

For completeness, the determinant identity used here follows by Schur-complement elimination of the central block, whose determinant is positive by (HCT2). The remaining 2×2 determinant is the product of its diagonal entries minus the product of its off-diagonal entries; multiplication by the central determinant squared gives (HCT25). For $n = 1$ the central block has size zero and determinant one, giving the same identity directly.

Put $\kappa_0^{(a)} = 0$ and, for $n \geq 1$,

$$\kappa_n^{(a)} = \frac{H_{n+1}^{(a)} H_{n-1}^{(a+2)}}{(H_n^{(a+1)})^2} > 0. \quad (\text{HCT26})$$

Then the exact logarithmic second difference is

$$\log H_n^{(a+2)} - 2 \log H_n^{(a+1)} + \log H_n^{(a)} = \log(1 + \kappa_n^{(a)}). \quad (\text{HCT27})$$

For $n \geq 1$ its equivalent positive Toda expression is

$$0 < \frac{d_n^{(a)}}{c_{n-1}^{(a+1)}} = \frac{\kappa_n^{(a)}}{1 + \kappa_n^{(a)}} < 1, \quad \log(1 + \kappa_n^{(a)}) = -\log\left(1 - \frac{d_n^{(a)}}{c_{n-1}^{(a+1)}}\right). \quad (\text{HCT28})$$

Substitution of (HCT9)–(HCT10), followed by (HCT25), proves every equality and inequality in (HCT28). In particular, if $R_n^{(a)} = H_n^{(a+1)}/H_n^{(a)}$, then

$$R_n^{(a)} = \prod_{j=0}^{n-1} c_j^{(a)}, \quad \frac{R_n^{(a+1)}}{R_n^{(a)}} = 1 + \kappa_n^{(a)}. \quad (\text{HCT29})$$

For a finite integer $l \geq 1$, exact telescoping gives

$$\log \frac{H_n^{(a+l+1)}/H_n^{(a+1)}}{H_n^{(a+l)}/H_n^{(a)}} = \sum_{s=0}^{l-1} \log(1 + \kappa_n^{(a+s)}). \quad (\text{HCT30})$$

This quantity is strictly positive when $n \geq 1$. The signed claim in (HCT27)–(HCT30) concerns a -differences of the indicated determinant ratios. The degree difference is the different exact identity $\log R_{n+1}^{(a)} - \log R_n^{(a)} = \log c_n^{(a)}$; no sign or asymptotic size of this last logarithm is needed.

HCT7. Return to both original full matrices. Retain the full matrices from the original real coordinate:

$$M_{a,r}[i,j] = \int_{\mathbb{R}} y^{2a+i+j} d\sigma(y) \quad (0 \leq i, j < r), \quad \mathcal{D}_{a,r} = \det M_{a,r}. \quad (\text{HCT31})$$

Let $n_0 = \lceil r/2 \rceil$, $n_1 = \lfloor r/2 \rfloor$. Permute the original indices to $0, 2, 4, \dots, 1, 3, 5, \dots$ using the same permutation in the source and its dual. Odd moments vanish by evenness of σ . The even-even block is $[\mu_{a+i+j}]_{i,j < n_0}$ and the odd-odd block is $[\mu_{a+1+i+j}]_{i,j < n_1}$. Thus the exact congruence and its determinant are

$$P_r M_{a,r} P_r^T = \text{diag}([\mu_{a+i+j}]_{i,j < n_0}, [\mu_{a+1+i+j}]_{i,j < n_1}), \quad \mathcal{D}_{a,r} = H_{n_0}^{(a)} H_{n_1}^{(a+1)}. \quad (\text{HCT32})$$

The determinant sign is $\det(P_r)^2 = 1$. At the function-space level the same map sends the coefficient vector of $p(t) + yq(t)$, with $t = y^2$, to the two coefficient vectors of (p, q) . Its exact norm identity is

$$\int_{\mathbb{R}} y^{2a} |p(y^2) + yq(y^2)|^2 d\sigma(y) = \int_0^\infty |p(t)|^2 d\nu_a(t) + \int_0^\infty |q(t)|^2 d\nu_{a+1}(t).$$

The cross term is odd. This proves the typed orthogonal direct-sum map with both original parity sectors retained.

For the even integer $q = e(k+1)^2 = 2n$ specified by the packet, the two original ranks are therefore exactly

$$\mathcal{D}_{a,q} = H_n^{(a)} H_n^{(a+1)}, \quad \mathcal{D}_{a,q+1} = H_{n+1}^{(a)} H_n^{(a+1)}. \quad (\text{HCT33})$$

Only the stipulated integrality and evenness of q are used here; the packet parameter e is left unchanged. In particular

$$\frac{\mathcal{D}_{a,q+1}}{\mathcal{D}_{a,q}} = h_n^{(a)}, \quad \frac{\mathcal{D}_{a+l,q+1}/\mathcal{D}_{a,q+1}}{\mathcal{D}_{a+l,q}/\mathcal{D}_{a,q}} = \prod_{s=0}^{l-1} c_n^{(a+s)}. \quad (\text{HCT34})$$

The empty product at $l = 0$ is one.

Write $Z_a = \mathcal{D}_{a,q} \mathcal{D}_{a,q+1}$ while keeping the two original ranks fixed. Equations (HCT29), (HCT33) give

$$Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2, \quad (\text{HCT35})$$

$$\frac{Z_{a+1}}{Z_a} = R_n^{(a)} R_{n+1}^{(a)} (R_n^{(a+1)})^2, \quad (\text{HCT36})$$

$$\frac{Z_{q+l}}{Z_q} = \prod_{s=0}^{l-1} \left\{ \left(\prod_{j=0}^{n-1} c_j^{(q+s)} \right) \left(\prod_{j=0}^n c_j^{(q+s)} \right) \left(\prod_{j=0}^{n-1} c_j^{(q+s+1)} \right)^2 \right\}. \quad (\text{HCT37})$$

Equation (HCT37) evaluates the paired determinant ratio at the requested original exponents $a = q, q + l$ by the positive rational evolution (HCT21)–(HCT23). Each of the four factors supplied by the two full matrices remains visible. The γ_0 factors in (HCT19)–(HCT24) cancel in this particular ratio because both numerator and denominator have the same total matrix dimension; they are present in the individual determinants throughout.

Finally the exact curvature of this same paired determinant is

$$\frac{Z_{a+2} Z_a}{Z_{a+1}^2} = (1 + \kappa_n^{(a)})(1 + \kappa_{n+1}^{(a)})(1 + \kappa_n^{(a+1)})^2 > 1. \quad (\text{HCT38})$$

For $l \geq 1$ the consequent finite shift identity is

$$\begin{aligned} \log \frac{Z_{a+l+1}/Z_{a+1}}{Z_{a+l}/Z_a} &= \sum_{s=0}^{l-1} [\log(1 + \kappa_n^{(a+s)}) + \log(1 + \kappa_{n+1}^{(a+s)}) \\ &\quad + 2\log(1 + \kappa_n^{(a+s+1)})] > 0. \end{aligned} \quad (\text{HCT39})$$

These statements follow by adding (HCT27) for exactly the four blocks in (HCT35), exponentiating for (HCT38), and telescoping for (HCT39); they make no replacement of q by an independent large parameter. They provide the full finite return, its positive contiguous increments, and its exact degree-step comparison. A growth rate along the packet values of q, k, l requires evaluating these explicit positive quantities; no such rate is assumed in any identity above.

10 The complete universal Gamma center is a positive rational product

Use exactly the all-index moments m_j in PHT2. The universal center identified in the low-endpoint transport calculation is the following finite quantity:

$$\begin{aligned} \mathcal{D}_{l,N} &= \prod_{a=0}^N \prod_{j=0}^{2l-1} (a + j + 1/2), \\ P_k &= \log \mathcal{D}_{l,q-1} + \log \mathcal{D}_{l,q} - \log \mathcal{D}_{l,2q-1} - \log \mathcal{D}_{l,2q}, \\ W_k &= P_k - \log \frac{m_{2q+2l}}{m_{2q}} + \log \frac{D_{q+l,q}}{D_{q,q}} + \log \frac{D_{q+l,q+1}}{D_{q,q+1}}. \end{aligned} \tag{PHT20}$$

Here $\mathcal{D}_{l,N}$ denotes exactly the source product called $D_{l,N}$ in GCR and LET, while $D_{a,r}$ retains its moment-determinant definition from PHT8. Define

$$G_{q,l} = \left(\prod_{a=q}^{2q-1} \prod_{j=0}^{2l-1} (a + j + 1/2) \right) \left(\prod_{a=q+1}^{2q} \prod_{j=0}^{2l-1} (a + j + 1/2) \right) > 0. \tag{PHT21}$$

Cancellation of the explicitly overlapping monic products gives $e^{P_k} = G_{q,l}^{-1}$. The low moment ratio retains its precise index conversion:

$$\frac{m_{2q+2l}}{m_{2q}} = \frac{\mu_{q+l}}{\mu_q} = \frac{H_1^{(q+l)}}{H_1^{(q)}} = \prod_{s=0}^{l-1} c_0^{(q+s)}. \tag{PHT22}$$

The last equality is HCT13 at degree zero and then finite telescoping. Combining PHT11, HCT37 and PHT22 proves

$$\begin{aligned} W_k &= \log \mathcal{W}_{q,l}, \\ \mathcal{W}_{q,l} &= \frac{1}{G_{q,l}} \prod_{s=0}^{l-1} \frac{\left(\prod_{j=0}^{n-1} c_j^{(q+s)} \right) \left(\prod_{j=0}^n c_j^{(q+s)} \right) \left(\prod_{j=0}^{n-1} c_j^{(q+s+1)} \right)^2}{c_0^{(q+s)}}, \quad n = q/2. \end{aligned} \tag{PHT23}$$

Every factor is positive and rational: the initial c, d values are the exact rational Gamma values HCT21, and HCT22–23 use only positive sums, products and quotients. The initial norms still contain c_σ , as HCT19–20 record. Their common powers cancel in PHT11 and PHT22 only after their full source determinants are constructed.

Thus PHT23 is a finite evaluation procedure for the entire universal center in PHT20, requiring neither a new moment quadrature nor a difference of nearly equal Gram determinants. One sufficient finite initial range is $0 \leq j \leq n + q + l$: HCT22–23 then compute all needed coefficients through shift $q + l$ with the proved decreasing-index range in HCT24. This statement specifies an exact finite procedure. It does not assert a bit-complexity estimate or an asymptotic value along the original coupled sequence $q = e(k+1)^2, l = (k-1)/4$. The positive contiguous identities HCT25–39 provide the exact equations for that next calculation, with all ranks and original source factors retained.

The actual growth calculation

11 The original four-product cancellation with a finite remainder

Retain $k = 4l + 1$, $l \geq 1$, $m \geq 1$, $e = 1 + k(m - 1)$ and $q = e(k + 1)^2 = 2n$. The symbols e and n here are the original packet multiplicity and the exact half of its degree. Define the original source products and their signed combination by

$$D_{l,N} = \prod_{a=0}^N \prod_{j=0}^{2l-1} (a + j + 1/2), \quad P_k = \log D_{l,q-1} + \log D_{l,q} - \log D_{l,2q-1} - \log D_{l,2q}. \quad (\text{WGP1})$$

The present calculation estimates these literal products, including both distinct endpoint ranges. In particular it does not replace the two ranges by a common interval and omit their endpoint correction.

Cancelling identical factors in the four specified finite products gives

$$P_k = - \sum_{j=0}^{2l-1} \left\{ \sum_{a=q}^{2q-1} \log(a + j + 1/2) + \sum_{a=q+1}^{2q} \log(a + j + 1/2) \right\}. \quad (\text{WGP2})$$

Put $x_b = (b + 1/2)/q$ for $q \leq b \leq 2q - 1$. The change of index $b = a - 1$ in the second sum retains its full endpoints and yields

$$\tilde{P}_{q,l} := P_k + 4lq \log q = - \sum_{b=q}^{2q-1} \sum_{j=0}^{2l-1} \{ \log(x_b + j/q) + \log(x_b + (j + 1)/q) \}. \quad (\text{WGP3})$$

This defines an explicitly recorded scalar correction to P_k ; the original products in WGP1 are still the objects to which it refers.

An elementary quadrature bound with its sign. For a twice continuously differentiable real F on $[1, 2]$, divide that interval into the q cells of width $h = 1/q$ with centres x_b . The identity

$$F(x_b + u) - F(x_b) - uF'(x_b) = \begin{cases} \int_0^u (u - v)F''(x_b + v) dv, & u \geq 0, \\ \int_u^0 (v - u)F''(x_b + v) dv, & u < 0 \end{cases}$$

follows by two integrations of F'' . Its absolute value is at most $\|F''\|_\infty u^2/2$. Integration over each symmetric cell kills the linear term, giving

$$\left| \sum_b F(x_b) - q \int_1^2 F(x) dx \right| \leq \frac{\|F''\|_\infty}{24q}. \quad (\text{WGP4})$$

The remainder has the sign of F'' when this sign is constant; hence the midpoint sum is below the integral multiplied by q for convex F and above it for concave F . Applying the formula to the three literal functions $\log x$, x^{-1} , x^{-2} gives

$$\begin{aligned} \sum_b \log x_b &= q(2 \log 2 - 1) + \varepsilon_0, & 0 \leq \varepsilon_0 &\leq (24q)^{-1}, \\ \sum_b x_b^{-1} &= q \log 2 - \varepsilon_1, & 0 \leq \varepsilon_1 &\leq (12q)^{-1}, \\ \sum_b x_b^{-2} &= q/2 - \varepsilon_2, & 0 \leq \varepsilon_2 &\leq (4q)^{-1}. \end{aligned} \quad (\text{WGP5})$$

Here the second derivative absolute bounds are respectively 1, 2, 6. The integrals are evaluated by their displayed elementary antiderivatives.

For every $u \geq 0$, integrating $1/(1+u) = 1 - u + u^2/(1+u)$ gives

$$\log(1+u) = u - u^2/2 + r(u), \quad r(u) = \int_0^u \frac{t^2}{1+t} dt, \quad 0 \leq r(u) \leq u^3/3. \quad (\text{WGP6})$$

Use this exact formula at $u = j/(qx_b)$ and $(j+1)/(qx_b)$. The finite sums of their numerators are

$$\begin{aligned} \sum_{j < 2l} (j + j + 1) &= 4l^2, \\ \sum_{j < 2l} (j^2 + (j+1)^2) &= (16l^3 + 2l)/3, \\ \sum_{j < 2l} (j^3 + (j+1)^3) &= 8l^4 + 2l^2. \end{aligned} \quad (\text{WGP7})$$

These identities follow by adding the two shifted ranges; they also follow directly from telescoping the differences of the powers j^2, j^3, j^4 . Consequently WGP3 is the exact identity

$$\tilde{P}_{q,l} = -4l \sum_b \log x_b - \frac{4l^2}{q} \sum_b x_b^{-1} + \frac{8l^3 + l}{3q^2} \sum_b x_b^{-2} - \mathcal{R}_{q,l}, \quad (\text{WGP8})$$

where the original sum of the $2(2l)q$ nonnegative remainders satisfies

$$0 \leq \mathcal{R}_{q,l} \leq \frac{8l^4 + 2l^2}{3q^3} \sum_b x_b^{-3} \leq \frac{8l^4 + 2l^2}{3q^2}. \quad (\text{WGP9})$$

The last inequality uses $x_b \geq 1$ and the exact number q of cells. There is no smallness hypothesis on j/q in WGP6–9.

Substituting WGP5 gives the finite expansion

$$\begin{aligned} P_k &= -4lq \log q - 4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} + \mathcal{E}_{q,l}, \\ -\mathcal{E}_{q,l}^- &\leq \mathcal{E}_{q,l} \leq \mathcal{E}_{q,l}^+, \\ \mathcal{E}_{q,l}^- &= \frac{l}{6q} + \frac{8l^3 + l}{12q^3} + \frac{8l^4 + 2l^2}{3q^2}, \quad \mathcal{E}_{q,l}^+ = \frac{l^2}{3q^2}. \end{aligned} \quad (\text{WGP10})$$

Indeed the exact error is $-4l\varepsilon_0 + (4l^2/q)\varepsilon_1 - (8l^3 + l)\varepsilon_2/(3q^2) - \mathcal{R}_{q,l}$. Every coefficient is positive before its displayed sign is applied; the inequalities in WGP10 follow separately from WGP5 and WGP9. Since $q \geq (4l+2)^2$, this error is $O(1)$ at fixed $m = 1$. For any fixed $m \geq 2$ it is $O(l^{-2})$ as $l \rightarrow \infty$, because $e = 1 + (4l+1)(m-1)$ and hence q is of order l^3 . In particular the explicit $-4l^2 \log 2$ term survives on the q scale at $m = 1$ and has been retained. The next term is also displayed with its sign and original endpoints rather than absorbed into that scale.

Exact interface to the positive original Gamma recurrence. The Gamma measure is still $d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi)$. For $a, j \geq 0$ let $H_j^{(a)}$ and $c_j^{(a)}$ have exactly their HCT definitions: $H_j^{(a)} = \det[\int y^{2(a+i+r)} d\sigma]_{i,r < j}$ and $c_j^{(a)} = H_{j+1}^{(a+1)} H_j^{(a)} / (H_{j+1}^{(a)} H_j^{(a+1)})$. All original mass factors are present in the four determinants in this expression. Define only the scalar notation $\lambda_j^{(a)} = \log(c_j^{(a)}/q^2)$, keeping the same q for every shift a . HCT37 and the exact low identity PHT22 give

$$\begin{aligned} W_k &= \tilde{P}_{q,l} + \sum_{s=0}^{l-1} \left[2 \sum_{j=0}^{n-1} \lambda_j^{(q+s)} + \lambda_n^{(q+s)} \right. \\ &\quad \left. + 2 \sum_{j=0}^{n-1} \lambda_j^{(q+s+1)} - \lambda_0^{(q+s)} \right], \quad n = q/2. \end{aligned} \quad (\text{WGP11})$$

To verify the entire scale cancellation, each bracket has $2n + 1 + 2n - 1 = 4n = 2q$ copies of $2 \log q$ before this scalar notation is introduced. Across the l original shifts this is $4lq \log q$, exactly the term recorded in WGP3. The two high ranks, the doubled odd block and the subtracted low endpoint remain visible in WGP11. The finite remainders in WGP10 therefore enter the same W_k as in LET19–21, with coefficient $+1$; in the arithmetic receiver $B_{\text{ar}} = B_0 + \delta_\sigma - \mathfrak{T}_k$ their signs reverse because the coefficient of W_k is -1 .

The constant-scale term obtained during the independent check. The following refinement retains the same source products and all their endpoints. Integrating $1/(1+u) = 1 - u + u^2 - u^3/(1+u)$ gives

$$\log(1+u) = u - u^2/2 + u^3/3 - r_4(u), \quad r_4(u) = \int_0^u \frac{t^3}{1+t} dt, \quad 0 \leq r_4(u) \leq u^4/4.$$

Write

$$A_3 = 8l^4 + 2l^2, \quad A_4 = \sum_{j < 2l} (j^4 + (j+1)^4) = \frac{192l^5 + 80l^3 - 2l}{15}. \quad (\text{WGP12})$$

To verify the last polynomial, set $N = 2l$ and subtract its value $2N^5/5 + 2N^3/3 - N/15$ at N from its value at $N + 1$. Expansion gives exactly $N^4 + (N + 1)^4$; at $N = 0$ both sides vanish. This proves the finite sum by induction. WGP4 at x^{-3} and the convex midpoint inequality at x^{-4} give

$$\sum_b x_b^{-3} = 3q/8 - \varepsilon_3, \quad 0 \leq \varepsilon_3 \leq (2q)^{-1}, \quad \sum_b x_b^{-4} \leq 7q/24.$$

The derivatives and integrals used here are $(x^{-3})'' = 12x^{-5}$, $\int_1^2 x^{-3} dx = 3/8$, and $\int_1^2 x^{-4} dx = 7/24$. Thus the nonnegative quartic remainder Q_4 obeys $0 \leq Q_4 \leq 7A_4/(96q^3)$, and the exact refinement is

$$\boxed{P_k = -4lq \log q - 4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} - \frac{l^4 + l^2/4}{q^2} + \mathcal{E}_{q,l}^{(3)},} \quad (\text{WGP13})$$

$$-\frac{l}{6q} - \frac{8l^3 + l}{12q^3} \leq \mathcal{E}_{q,l}^{(3)} \leq \frac{l^2}{3q^2} + \frac{A_3}{6q^4} + \frac{7A_4}{96q^3}.$$

Indeed its exact error is

$$\mathcal{E}_{q,l}^{(3)} = -4l\varepsilon_0 + \frac{4l^2}{q}\varepsilon_1 - \frac{8l^3 + l}{3q^2}\varepsilon_2 + \frac{A_3}{3q^3}\varepsilon_3 + Q_4. \quad (\text{WGP14})$$

The cubic term has coefficient $-A_3(3q/8)/(3q^3) = -(l^4 + l^2/4)/q^2$, retaining its negative sign. The displayed bounds give $\mathcal{E}_{q,l}^{(3)} = O(l^{-1})$ for $m = 1$ and $O(l^{-2})$ for every fixed $m \geq 2$. For $m = 1$ the earlier exact remainder therefore satisfies $\mathcal{E}_{q,l} = -1/256 + O(l^{-1})$, because $q = (4l + 2)^2$. WGP13 supersedes the coarser numerical expansion WGP10 when this constant-scale information is used; WGP10 remains a valid independently proved enclosure. Both versions enter the identical WGP11 and LET21 maps, with their already specified opposite arithmetic sign.

12 A positive growing-degree bound for the complete Gamma return

Retain exactly the measure, monic polynomials, norms and Christoffel coefficients of HCT1–24:

$$\begin{aligned} d\sigma(y) &= \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \\ \nu &= (y \mapsto y^2)_* \sigma, \quad d\nu_a(t) = t^a d\nu(t), \\ h_j^{(a)} &= \int_0^\infty (P_j^{(a)}(t))^2 d\nu_a(t), \quad c_j^{(a)} = \frac{h_j^{(a+1)}}{h_j^{(a)}}. \end{aligned} \tag{RWB1}$$

Every $P_j^{(a)}$ is monic of degree j . The original mass $\sqrt{2\pi}$ remains in the norms; the last quotient is the exact HCT13 identity. Throughout this section a, j are nonnegative integers, and the estimates containing $4a - 1$ require $a \geq 1$.

The logarithmic derivative of the original Gamma density. Write $\sigma(y)$ also for the displayed density and set $L(y) = -\sigma'(y)/\sigma(y)$. For every $y \in \mathbb{R}$,

$$-\frac{1}{2} \leq yL(y) - \frac{\pi}{2}|y| \leq \frac{3}{2}, \quad \left| y \left(L(y) - \frac{\pi}{2} \operatorname{sgn} y \right) \right| \leq \frac{3}{2}. \tag{RWB2}$$

The expressions at $y = 0$ have their continuous value zero. Here is a derivation retaining the exact parameter $1/4$. For $\Re z > 0$, integration by parts in the beta integral and the change $u = Nx$ give

$$\frac{N!N^z}{z(z+1)\cdots(z+N)} = \int_0^N u^{z-1}(1-u/N)^N du \longrightarrow \Gamma(z). \tag{RWB3}$$

The inequality $(1-u/N)^N \leq e^{-u}$ proves locally uniform convergence on every compact subset of $\Re z > 0$: near zero use the least real part on that compact set, and above one use its greatest real part. The same majorants with a factor $|\log u|$ justify differentiated convergence. Nonvanishing at each $z = \alpha + iv$, $\alpha > 0$, follows directly from the same limits, since

$$\frac{|\Gamma(\alpha + iv)|}{\Gamma(\alpha)} = \lim_{N \rightarrow \infty} \prod_{r=0}^N \left(1 + \frac{v^2}{(r+\alpha)^2} \right)^{-1/2} > 0.$$

The infinite product has a positive limit because $\sum_r \log(1+v^2/(r+\alpha)^2)$ converges by comparison with $\sum_r v^2/(r+\alpha)^2$. Thus logarithmic differentiation, with $z = 1/4 + iv$, gives the convergent imaginary-part series

$$\operatorname{Im} \frac{\Gamma'(1/4 + iv)}{\Gamma(1/4 + iv)} = \sum_{r=0}^{\infty} \frac{v}{(r+1/4)^2 + v^2}, \quad L(y) = \operatorname{Im} \frac{\Gamma'(1/4 + iy/2)}{\Gamma(1/4 + iy/2)}. \tag{RWB4}$$

Only an imaginary part is taken, so the real $\log N$ term disappears. For $v > 0$ let $F_v(x) = v^2/(x^2 + v^2)$, a positive decreasing function on $[0, \infty)$. Integral comparison on each unit interval proves

$$\int_{1/4}^{\infty} F_v(x) dx \leq \sum_{r \geq 0} F_v(r + 1/4) \leq F_v(1/4) + \int_{1/4}^{\infty} F_v(x) dx.$$

Since $\int_0^\infty F_v(x) dx = \pi v/2$, the difference between the sum and this last integral is at least $-1/4$ and at most

$$F_v(1/4) - \int_0^{1/4} F_v(x) dx \leq \frac{3}{4} F_v(1/4) \leq \frac{3}{4}.$$

For $y = 2v > 0$, RWB4 gives $yL(y) = 2 \sum_r F_v(r + 1/4)$. Multiplying the preceding interval by two proves RWB2 at positive y . Evenness of σ makes $yL(y)$ even, proving the assertion on the whole original real line.

All integrations by parts below are justified by the exponential bound proved from the Euler contour in HCT1. Formula RWB4 shows that L is continuous at zero; RWB2 bounds it at infinity, so the differentiated integrands have the required integrability as well.

An inverse-square estimate with its exact polynomial space. For $a \geq 1$ fix $Q = P_j^{(a+1)}$ and put

$$J_{a,j} = \frac{\int_0^\infty Q(t)^2 d\nu_a(t)}{\int_0^\infty Q(t)^2 d\nu_{a+1}(t)}.$$

Then

$$0 < J_{a,j} \leq \left(\frac{\pi}{4a-1} \right)^2. \quad (\text{RWB5})$$

To prove this, the exact density of ν_{a+1} from HCT1 is $t^{a+1/2}\sigma(\sqrt{t})$ on $(0, \infty)$. Integrate the derivative of $Q(t)^2 t^{a+1/2}\sigma(\sqrt{t})$ over that interval. Its boundary values vanish: at zero $a+1/2 > 0$, and at infinity the factor has a polynomial times exponential bound. The term $2QQ'$ integrates to zero because $\deg Q' < j$ and Q is orthogonal in the exact measure ν_{a+1} . Dividing by $h_j^{(a+1)}$ therefore gives

$$(a+1/2)J_{a,j} = \frac{1}{2} \frac{\int_0^\infty Q(t)^2 L(\sqrt{t})/\sqrt{t} d\nu_{a+1}(t)}{h_j^{(a+1)}}. \quad (\text{RWB6})$$

RWB2 implies $L(\sqrt{t})/\sqrt{t} \leq \pi/(2\sqrt{t}) + 3/(2t)$. Cauchy-Schwarz in the same positive measure gives an upper bound $\sqrt{J_{a,j}}$ for the ratio obtained by inserting $t^{-1/2}$. Consequently

$$(a-1/4)J_{a,j} \leq \frac{\pi}{4} \sqrt{J_{a,j}}.$$

The denominator is positive and $J_{a,j} > 0$, so division proves RWB5. No assertion concerning an unspecified inverse operator is used: both integrals defining $J_{a,j}$ are the original polynomial integrals.

A lower bound including both the degree and the shift. For every $a \geq 1, j \geq 0$,

$$\sqrt{c_j^{(a)}} \geq \frac{2(4a-1)(2a+2j+1)}{\pi(4a+2)} = \frac{4}{\pi} (a+j+1/2) \frac{4a-1}{4a+2}. \quad (\text{RWB7})$$

Here the square root is the positive one. Work in the original real-line measure $dw_a(y) = y^{2a} d\sigma(y)$ and define

$$R(y) = P_j^{(a)}(y^2), \quad O(y) = yP_j^{(a+1)}(y^2). \quad (\text{RWB8})$$

Evenness of w_a and the defining orthogonality on the square pushforward show that R, O are the monic orthogonal polynomials of degrees $2j, 2j+1$ in $L^2(w_a)$. Their norms are exactly $h_j^{(a)}$ and $h_j^{(a+1)}$; both parity sectors have been retained. In particular

$$\int O'R dw_a = (2j+1)h_j^{(a)}, \quad \int OR' dw_a = 0, \quad \int \frac{OR}{y} dw_a = h_j^{(a)}. \quad (\text{RWB9})$$

The last integrand is a polynomial: $O/y = P_j^{(a+1)}(y^2)$ is monic of the same degree as R , so their difference has lower degree. Integrating $(ORy^{2a}\sigma)'$ and using RWB9 proves the exact identity

$$2a+2j+1 = \frac{\int ORL dw_a}{h_j^{(a)}}. \quad (\text{RWB10})$$

At zero the boundary term is zero because O is odd, and the apparent $1/y$ term has just been exhibited as a polynomial. At both infinite ends exponential decay proves vanishing.

Write $L(y) = (\pi/2) \operatorname{sgn} y + \rho(y)$. Cauchy-Schwarz bounds the first contribution in RWB10 by $(\pi/2)\sqrt{c_j^{(a)}}$. For the second, $|y\rho(y)| \leq 3/2$ from RWB2 and $O/y = P_j^{(a+1)}(y^2)$ give the bound

$$\frac{3}{2} \sqrt{\frac{\int (P_j^{(a+1)}(y^2))^2 dw_a}{h_j^{(a)}}} = \frac{3}{2} \sqrt{c_j^{(a)} J_{a,j}} \leq \frac{3\pi}{2(4a-1)} \sqrt{c_j^{(a)}}.$$

Thus $2a+2j+1 \leq (\pi/2)(4a+2)/(4a-1)\sqrt{c_j^{(a)}}$, which is RWB7, including its original degree and constants.

The low coefficient with the sign needed in the return. For $p \geq 0$ write $I_p = \int_0^\infty y^p \sigma(y) dy > 0$. Integration of $(y^{p+1} \sigma(y))'$ and RWB2 give

$$\frac{2p-1}{\pi} \leq \frac{I_{p+1}}{I_p} \leq \frac{2p+3}{\pi}. \quad (\text{RWB11})$$

For $p < 1/2$ the lower bound is simply a valid negative bound; it will not be used there. Indeed the integrated identity is $(p+1)I_p = \int y^{p+1} L(y) \sigma(y) dy$; substituting the interval in RWB2 proves both signs in RWB11. Since $P_0^{(a)} = 1$, evenness gives $c_0^{(a)} = I_{2a+2}/I_{2a}$. Multiplying two upper bounds in RWB11 proves

$$c_0^{(a)} \leq \frac{(4a+3)(4a+5)}{\pi^2}. \quad (\text{RWB12})$$

This is the upper bound appropriate to the negative low-endpoint term; RWB7 will be used only in the positively signed high terms.

The full four-endpoint functional at the original ranks. Retain the packet integers

$$k = 4l + 1, \quad l \geq 1, \quad m \geq 1, \quad e = 1 + k(m-1), \quad q = e(k+1)^2 = 2n. \quad (\text{RWB13})$$

Let W_k be exactly LET19. With

$$G = \prod_{b=0}^{2l-1} \left\{ \prod_{u=q}^{2q-1} (u+b+1/2) \prod_{u=q+1}^{2q} (u+b+1/2) \right\},$$

HCT37 and the low moment ratio give the exact identity

$$e^{W_k} = \frac{1}{G} \prod_{s=0}^{l-1} \frac{\left(\prod_{j=0}^{n-1} c_j^{(q+s)} \right)^2 c_n^{(q+s)} \left(\prod_{j=0}^{n-1} c_j^{(q+s+1)} \right)^2}{c_0^{(q+s)}}. \quad (\text{RWB14})$$

Here $P_k = -\log G$ follows by cancelling the overlapping literal products $D_{l,N}$ in LET19, with both endpoint intervals as displayed. The source masses cancel only in these prescribed equal-dimension ratios. There are $4lq$ original linear factors in G . In each s factor the net number of c coefficients is $2q$. Therefore division of each linear factor by q and each c by q^2 cancels exactly $4lq \log q$ on the two sides and yields

$$\begin{aligned} W_k = & - \sum_{b=0}^{2l-1} \left[\sum_{u=q}^{2q-1} \log \frac{u+b+1/2}{q} + \sum_{u=q+1}^{2q} \log \frac{u+b+1/2}{q} \right] \\ & + \sum_{s=0}^{l-1} \left[2 \sum_{j=0}^{n-1} \log \frac{c_j^{(q+s)}}{q^2} + \log \frac{c_n^{(q+s)}}{q^2} + 2 \sum_{j=0}^{n-1} \log \frac{c_j^{(q+s+1)}}{q^2} - \log \frac{c_0^{(q+s)}}{q^2} \right]. \end{aligned} \quad (\text{RWB15})$$

This is an exact equality of the original quantities, with every scale factor accounted for, rather than replacement of the original objects.

Define $r_q = (4q-1)/(4q+2)$. The function $a \mapsto (4a-1)/(4a+2)$ is increasing for $a \geq 1$. RWB7 thus implies, for both $a = q+s$ and $a = q+s+1$,

$$\log \frac{c_j^{(a)}}{q^2} \geq 2 \log \frac{4}{\pi} + 2 \log(1+j/q) + 2 \log r_q. \quad (\text{RWB16})$$

RWB12 and $s \leq l-1$ give the simultaneous upper bound

$$\frac{c_0^{(q+s)}}{q^2} \leq U_{q,l} := \frac{(4q+4l-1)(4q+4l+1)}{\pi^2 q^2} \leq \left(\frac{4}{\pi} \right)^2 \left(1 + \frac{l+1/4}{q} \right)^2. \quad (\text{RWB17})$$

For the increasing function $x \mapsto \log(1+x)$, comparison of its left and right sums on $[0, 1/2]$ gives

$$\sum_{j=0}^{n-1} \log(1+j/q) \geq q \int_0^{1/2} \log(1+x) dx - \log(3/2) = q \left[\frac{3}{2} \log(3/2) - \frac{1}{2} \right] - \log(3/2). \quad (\text{RWB18})$$

Inserting RWB16–18 into the s bracket of RWB15, counting its $2q+1$ positive coefficients, gives the lower bound

$$4q [\log(4/\pi) + 3 \log(3/2) - 1] + 2 \log(4/\pi) + (4q+2) \log r_q - 6 \log(3/2) - \log U_{q,l}. \quad (\text{RWB19})$$

For the first line of RWB15 use $\log((u+b+1/2)/q) \leq \log(u/q) + (b+1/2)/u \leq \log(u/q) + (b+1/2)/q$. The sum of the latter corrections over both u intervals and all b is exactly $4l^2$. Concavity of $\log x$ shows that its composite trapezoidal sum on $[1, 2]$ is at most its integral: each chord lies below the graph. Consequently

$$\sum_{u=q}^{2q-1} \log(u/q) + \sum_{u=q+1}^{2q} \log(u/q) \leq 2q \int_1^2 \log x dx = 2q(2 \log 2 - 1).$$

The first line of RWB15 is therefore at least $-4lq(2 \log 2 - 1) - 4l^2$. Add l copies of RWB19 and use

$$(4q+2) \log r_q \geq -\frac{3(4q+2)}{4q-1} = -3 - \frac{9}{4q-1}, \quad \log(1+x) \leq x \quad (x \geq 0).$$

The first inequality follows by integrating $1/(1-t) \leq 1/(1-x)$ on $0 \leq t \leq x < 1$. This proves the completely explicit bound

$$W_k \geq 4lq \log \frac{27}{8\pi} - 4l^2 - 6l \log(3/2) - 3l - \frac{9l}{4q-1} - \frac{2l(l+1/4)}{q}. \quad (\text{RWB20})$$

Every term is at the original finite q, l , including the negatively signed low coefficient. In particular no large-rank value of a Hankel determinant or a recurrence coefficient has been assumed.

Uniform positivity and the actual packet limit. The preceding finite bound implies

$$W_k \geq \frac{lq}{64} > 0 \quad \text{for every } l \geq 1, m \geq 1. \quad (\text{RWB21})$$

Here are rational verifications of the constants. The identity

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx > 0$$

follows by polynomial division and $\int_0^1 (1+x^2)^{-1} dx = \pi/4$. Thus $27/(8\pi) > 189/176$. For $x \geq 0$, differentiation proves $\log(1+x) \geq 2x/(2+x)$, and hence $4 \log(27/(8\pi)) > 104/365$. Integrating $1/(1+t) \leq 1-t+t^2$ over $0 \leq t \leq 1/2$ gives $\log(3/2) \leq 5/12$. RWB13 gives $q \geq (4l+2)^2 \geq 36l$; the second inequality is $(4l+2)^2 - 36l = 4(4l-1)(l-1) \geq 0$. Dividing RWB20 by lq now yields

$$\frac{W_k}{lq} \geq \frac{104}{365} - \frac{1}{9} - \frac{11}{72} - \frac{1}{572} - \frac{5}{2592} = \frac{2349349}{135289440} = \frac{1}{64} + \frac{470903}{270578880} > \frac{1}{64}. \quad (\text{RWB22})$$

The four subtracted rational terms bound respectively $4l/q$, $(6 \log(3/2) + 3)/q$, $9/[q(4q-1)]$, and $2(l+1/4)/q^2$. This proves RWB21 on the entire stated packet domain.

Finally, along the original $k = 4l+1 \rightarrow \infty$ and any integer choices $m = m(l) \geq 1$, RWB13 gives $l/q \rightarrow 0$ and $1/q \rightarrow 0$. The exact finite remainder in RWB20 therefore proves

$$\liminf_{l \rightarrow \infty} \frac{W_{4l+1}}{lq} \geq 4 \log \frac{27}{8\pi} > 0. \quad (\text{RWB23})$$

This is a lower bound, with its finite remainder RWB20. No equality or numerical asymptotic value is asserted. The exact original receiving identities LET20–21 may now use this bound with their already proved signs and errors; the arithmetic contribution in those identities is left exactly as defined there.

A matching order bound from the original polynomial operator. For completeness, the exact original real-line recurrence HCT6 gives, for $a \geq 1, j \geq 0$ and $D = a + 2j$,

$$c_j^{(a)} \leq \left[\sqrt{(D+1)(D+1/2)} + \sqrt{D(D-1/2)} \right]^2 \leq (2a + 4j + 1/2)^2. \quad (\text{RWB24})$$

Indeed the norm of a monic orthogonal polynomial is the minimum norm among all monic polynomials of that degree: subtracting the orthogonal polynomial leaves a lower-degree polynomial perpendicular to it. Consequently

$$h_j^{(a+1)} \leq \int_0^\infty t(P_j^{(a)}(t))^2 d\nu_a(t) = \|yF\|_{L^2(\sigma)}^2, \quad F(y) = y^a P_j^{(a)}(y^2), \quad \|F\|_{L^2(\sigma)}^2 = h_j^{(a)}.$$

This is the explicit multiplication map on the original polynomials of degree at most D , with image degree at most $D + 1$. For the exact orthonormal basis $e_d = p_d / \sqrt{\gamma_0 d! (1/2)_d}$ from HCT5, its raising and lowering parts have coefficients $\sqrt{(d+1)(d+1/2)}$ and $\sqrt{d(d-1/2)}$, respectively. Their norms on degrees at most D are at most the two terms in RWB24; this follows by summing their mutually orthogonal images separately. The triangle inequality proves the first estimate. Since $D \geq 1$, the arithmetic–geometric mean bounds on the two square roots are $D + 3/4$ and $D - 1/4$, proving the second.

Combining the two lower bounds in RWB11, at $p = 2a, 2a + 1$, also proves

$$c_0^{(a)} \geq \frac{(4a-1)(4a+1)}{\pi^2} \quad (a \geq 1). \quad (\text{RWB25})$$

At the indices in RWB15, $a \leq q + l$ and $j \leq q/2$, so RWB24 gives $c_j^{(a)}/q^2 \leq [4 + (2l + 1/2)/q]^2$. RWB25 gives $c_0^{(q+s)}/q^2 \geq (16 - q^{-2})/\pi^2 > 1$, using $q \geq 36$ and $\pi < 22/7$. Every logarithm in the first line of RWB15 is positive. Hence

$$W_k \leq l(4q + 2) \log \left(4 + \frac{2l + 1/2}{q} \right) \leq 6lq. \quad (\text{RWB26})$$

For the last numerical bound, $q \geq 36l$ gives $4 + (2l + 1/2)/q \leq 293/72$. The positive exponential series gives

$$e^{10/7} \geq \sum_{r=0}^4 \frac{(10/7)^r}{r!} = \frac{29593}{7203} > \frac{293}{72},$$

where the rational inequality follows by cross multiplication. Thus the logarithm is less than $10/7$ and $(4q + 2)10/7 \leq 6q$ for $q \geq 10$. Together RWB21 and RWB26 prove the finite two-sided order estimate

$$\boxed{\frac{lq}{64} \leq W_k \leq 6lq \quad (l \geq 1, m \geq 1).} \quad (\text{RWB27})$$

The narrower lower remainder RWB20, rather than the coarse lower constant in RWB27, remains available for propagation into each original receiving formula.

Dictionary for the following auxiliary integral calculation. The real elliptic modulus denoted k within EIQ is the κ of GEL: $\kappa = k_{\text{EIQ}} \in (0, 1)$. EIQ's midpoint m is $m_{\text{EIQ}} = (u^2 + v^2)/2$. The original packet still has $k_{\text{packet}} = 4l + 1$ and integer multiplicity m_{packet} . The ensemble map GEL3 supplies parameters $\alpha = a/n$ and $\beta = \pi q/(2n)$; its original coupled sequences converge to $(\alpha, \beta) = (2, \pi)$. This is the explicit parameter dictionary connecting the next two sections to the original packet and source.

13 Exact equilibrium for the square-root logarithmic potential

Throughout this section $\alpha > 0$ and $\beta > 0$. On $(0, \infty)$ set

$$V_{\alpha, \beta}(x) = \beta\sqrt{x} - \alpha \log x, \quad \mathcal{E}_{\alpha, \beta}(\mu) = \int V_{\alpha, \beta}(x) d\mu(x) - \iint \log|x - y| d\mu(x) d\mu(y). \quad (\text{EIQ1})$$

The measures considered initially are probability measures for which $\int V_{\alpha, \beta} d\mu$ is finite. Because this potential is bounded below and tends to $+\infty$ at both ends, this implies $\int (\sqrt{x} + |\log x|) d\mu < \infty$. The positive part of $\log|x - y|$ is then integrable, by $\log^+|x - y| \leq \log(1 + x) + \log(1 + y)$. Thus the energy is well defined with value in $\mathbb{R} \cup \{+\infty\}$. All comparisons below also apply to the usual extended energy defined by integrating the kernel $\frac{1}{2}V_{\alpha, \beta}(x) + \frac{1}{2}V_{\alpha, \beta}(y) - \log|x - y|$: the kernel is bounded below, and finite energy for this kernel implies the same moment conditions. To see the latter assertion without cancellation, put $g(x) = \frac{1}{2}V_{\alpha, \beta}(x) - \log(1 + x)$. The kernel is at least $g(x) + g(y)$; g is bounded below and controls positive constant multiples of both $\sqrt{x}$ near infinity and $|\log x|$ near zero.

13.1 Existence and uniqueness of the two endpoints

For $0 < k < 1$ define the complete elliptic integrals by their actual real integrals

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta. \quad (\text{EIQ2})$$

There is exactly one $k \in (0, 1)$ satisfying

$$\frac{\sqrt{1 - k^2} K(k)}{E(k)} = \frac{\alpha}{\alpha + 2}. \quad (\text{EIQ3})$$

For this k put

$$r = \sqrt{1 - k^2}, \quad v = \frac{(\alpha + 2)\pi}{\beta E(k)}, \quad u = rv, \quad a = u^2, \quad b = v^2. \quad (\text{EIQ4})$$

In particular $0 < a < b < \infty$, and the endpoint equations, with all constants retained, are

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha + 2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}. \quad (\text{EIQ5})$$

Here is a proof of the assertion of uniqueness. Differentiation of the integrals in (EIQ2), and integration of the derivative of $\sin \theta \cos \theta / \sqrt{1 - k^2 \sin^2 \theta}$ between its two zero endpoint values, give

$$E'(k) = \frac{E(k) - K(k)}{k}, \quad K'(k) = \frac{E(k)}{k(1 - k^2)} - \frac{K(k)}{k}.$$

For completeness the second identity is equivalent, after differentiation under the integral sign, to

$$\int_0^{\pi/2} \left\{ \frac{k^2 \sin^2 \theta}{(1 - k^2 \sin^2 \theta)^{3/2}} - \frac{1 - k^2 \sin^2 \theta}{(1 - k^2) \sqrt{1 - k^2 \sin^2 \theta}} + \frac{1}{\sqrt{1 - k^2 \sin^2 \theta}} \right\} d\theta = 0;$$

the expression in braces is $-k^2(1-k^2)^{-1} d(\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta})/d\theta$. With $t = E(k)/K(k)$, the integral definitions show strictly $r^2 < t < 1$. Consequently, if $f(k) = rK(k)/E(k)$, then

$$\frac{kf'(k)}{f(k)} = \frac{t}{r^2} + \frac{1}{t} - 1 - \frac{1}{r^2} = \frac{(t-1)(t-r^2)}{r^2 t} < 0. \quad (\text{EIQ6})$$

At $k = 0$ the continuous limiting value is $f(0) = 1$. As $k \uparrow 1$, the integrand of $rK(k)$ tends to zero at every $\theta < \pi/2$ and is bounded by 1, while $E(k) \rightarrow 1$ by dominated convergence. Hence $f(k) \rightarrow 0$. This proves (EIQ3), including existence. It also proves smooth dependence of k, u, v, a, b on (α, β) in $(0, \infty)^2$, by the nonzero derivative in (EIQ6).

Let $m = (a+b)/2$, $R_s = \sqrt{(a+s)(b+s)}$ for $s \geq 0$, and $R_0 = uv$. Substitution $t = a \cos^2 \theta + b \sin^2 \theta$ gives, directly from (EIQ5),

$$\int_a^b \frac{V'_{\alpha,\beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 0, \quad \int_a^b \frac{tV'_{\alpha,\beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 2\pi. \quad (\text{EIQ7})$$

Indeed the three integrals with numerators $t^{-1/2}, t^{-1}, t^{1/2}$ are respectively $2K(k)/v$, $\pi/(uv)$, and $2vE(k)$, and $V'(t) = \beta/(2\sqrt{t}) - \alpha/t$.

13.2 A positive density and its exact Cauchy transform

The elementary Stieltjes integral, obtained by $s = xz^2$, is

$$\int_0^\infty \frac{ds}{\sqrt{s}(x+s)} = \frac{\pi}{\sqrt{x}} \quad (x > 0). \quad (\text{EIQ8})$$

Using this identity in (EIQ7) and applying Tonelli to its positive constituents gives

$$\frac{\alpha}{R_0} = \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}R_s}, \quad \int_0^\infty \frac{1-s/R_s}{\sqrt{s}} ds = 2vE(k). \quad (\text{EIQ9})$$

Both integrals converge; their integrands are $O(s^{-1/2})$ at zero and $O(s^{-3/2})$ at infinity. The second formula also follows by averaging $\sqrt{t} = \pi^{-1} \int_0^\infty s^{-1/2} t/(t+s) ds$ against the arcsine probability measure on $[a, b]$.

Define for every $x > 0$

$$\begin{aligned} h(x) &= \frac{\alpha}{xR_0} - \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}(x+s)R_s} \\ &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x+s)R_s} ds > 0. \end{aligned} \quad (\text{EIQ10})$$

The second equality is exactly (EIQ9), with no change of the potential. All integrals in (EIQ10) converge absolutely and locally uniformly in $x > 0$. The candidate density on the original positive half-line is

$$\rho_{\alpha,\beta}(x) = \begin{cases} \frac{\sqrt{(b-x)(x-a)}}{2\pi} h(x), & a < x < b, \\ 0, & x \notin (a, b). \end{cases} \quad (\text{EIQ11})$$

It is strictly positive on (a, b) and vanishes continuously with the square-root factor at each endpoint. Positivity here follows from the last integral in (EIQ10), not from a one-interval ansatz.

The following elementary transform computes its mass and resolvent. Choose

$$R(z) = \sqrt{(z-a)(z-b)}$$

on $\mathbb{C} \setminus [a, b]$ with $R(z)/z \rightarrow 1$ at infinity. Thus $R(x) > 0$ for $x > b$, $R(x) < 0$ for $x < a$, and $R(x+i0) = i\sqrt{(b-x)(x-a)}$ on (a, b) . Direct substitution $t = m + (b-a) \cos \theta/2$, followed by $w = \tan(\theta/2)$, proves

$$\frac{1}{\pi} \int_a^b \frac{dt}{(z-t)\sqrt{(b-t)(t-a)}} = \frac{1}{R(z)}.$$

For real $z > b$, the substitution reduces the integral to $\pi^{-1} \int_0^\infty 2 dw / [(z-b) + (z-a)w^2]$; the identity elsewhere follows by analytic continuation. Polynomial division then gives

$$\frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{z-t} dt = z - m - R(z), \quad \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{t+s} dt = m + s - R_s. \quad (\text{EIQ12})$$

For example the first identity follows from $(b-t)(t-a) = -(t-z)^2 + 2(m-z)(t-z) - (z-a)(z-b)$, the preceding arcsine transform, and the arcsine first moment m . Partial fractions in these two identities yield

$$H_s(z) := \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{(t+s)(z-t)} dt = 1 - \frac{R_s + R(z)}{z+s}. \quad (\text{EIQ13})$$

The formula at $z = -s$ is interpreted by its removable limit.

The mass of (EIQ11), computed using (EIQ12), is

$$\begin{aligned} \int_a^b \rho_{\alpha,\beta}(t) dt &= \frac{\alpha}{2R_0}(m - R_0) - \frac{\beta}{4\pi} \int_0^\infty \frac{(m+s)/R_s - 1}{\sqrt{s}} ds \\ &= -\frac{\alpha}{2} + \frac{\beta}{4\pi} \int_0^\infty \frac{1-s/R_s}{\sqrt{s}} ds = -\frac{\alpha}{2} + \frac{\beta v E(k)}{2\pi} = 1. \end{aligned} \quad (\text{EIQ14})$$

In particular this is a probability density.

Define its Cauchy transform with the explicit sign convention $G(z) = \int_a^b \rho_{\alpha,\beta}(t)/(z-t) dt$. For z outside both $[a, b]$ and $(-\infty, 0]$, formulas (EIQ10) and (EIQ13) give

$$\begin{aligned} G(z) &= \frac{\alpha}{2R_0} H_0(z) - \frac{\beta}{4\pi} \int_0^\infty \frac{H_s(z)}{\sqrt{s} R_s} ds \\ &= -\frac{\alpha}{2z} + \frac{\beta}{4\sqrt{z}} - \frac{R(z)h(z)}{2} = \frac{V'_{\alpha,\beta}(z) - R(z)h(z)}{2}. \end{aligned} \quad (\text{EIQ15})$$

Here the positive square root is continued with its negative-real-axis cut, and $h(z)$ is the first integral formula of (EIQ10). Every integral exchange is absolutely convergent, uniformly on compact subsets of this domain: near zero the auxiliary integrand is $O(s^{-1/2})$, and near infinity the original integral defining H_s is $O(s^{-1})$. The constant terms on the second line cancel by (EIQ9), and (EIQ8) supplies the term $\beta/(4\sqrt{z})$.

13.3 The Euler equality, its strict exterior inequality, and uniqueness

Set

$$U(x) = V_{\alpha,\beta}(x) - 2 \int_a^b \log |x-t| \rho_{\alpha,\beta}(t) dt \quad (x > 0). \quad (\text{EIQ16})$$

This function is continuous on $(0, \infty)$. The integral is continuous because the density is bounded, compactly supported, and the integral of $|\log |x-t||$ over an interval of length ε tends to zero uniformly in its centre. In the interior of $[a, b]$, differentiation of the logarithmic potential is the principal-value integral $\text{PV} \int \rho(t)/(x-t) dt$. This follows by subtracting $\rho(x)$ on a symmetric interval about x ; the remaining numerator is $O(|t-x|)$ because the density is continuously differentiable locally in the interior. The real part of (EIQ15) on the upper edge of the interval is precisely this principal value. Indeed subtracting $\rho(x)$ in $\int \rho(t)/(x+i\varepsilon-t) dt$ gives the same limit, while the constant term has real part equal to its elementary principal value. Since $R(x+i0)$ is purely imaginary and $h(x)$ is real, (EIQ15) proves $U'(x) = 0$ on (a, b) . There is therefore a real constant ℓ such that

$$U(x) = \ell \quad (a \leq x \leq b). \quad (\text{EIQ17})$$

For positive x outside this interval, (EIQ15) instead gives the exact real derivative

$$U'(x) = R(x)h(x) \begin{cases} < 0, & 0 < x < a, \\ > 0, & x > b. \end{cases} \quad (\text{EIQ18})$$

Combining (EIQ17), continuity, and these signs proves

$$U(x) > \ell \quad \text{for } x \in (0, a) \cup (b, \infty). \quad (\text{EIQ19})$$

This verifies the entire exterior inequality on the original domain, including the interval next to zero.

Here is a complete positivity argument for the energy of the difference of two measures. If ν is a finite signed real measure of total mass zero for which $|\log|x - y||$ is integrable against $|\nu| \otimes |\nu|$, the scalar identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt \quad (d > 0) \quad (\text{EIQ20})$$

follows by differentiating in d and evaluating at $d = 1$. One half of the integral over any finite subinterval of $(0, \infty)$ has the same sign as $-\log d$ and absolute value at most $|\log d|$. Consequently dominated convergence and the zero total mass give

$$-\iint \log|x - y| d\nu(x) d\nu(y) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t(x-y)^2} d\nu(x) d\nu(y) dt \geq 0. \quad (\text{EIQ21})$$

The nonnegative sign is proved, rather than assumed, by the Gaussian Fourier identity

$$\iint e^{-t(x-y)^2} d\nu(x) d\nu(y) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\xi^2/(4t)} |\widehat{\nu}(\xi)|^2 d\xi, \quad \widehat{\nu}(\xi) = \int e^{i\xi x} d\nu(x). \quad (\text{EIQ22})$$

The scalar Gaussian Fourier identity follows by completing the square in the Gaussian integral (or by differentiating its Fourier transform and its value at zero); its integration against the finite measures is justified by absolute integrability. Equality in (EIQ21) forces $\widehat{\nu} = 0$: the right side of (EIQ22) is strictly positive for every $t > 0$ if this continuous Fourier transform has a nonzero value. Finally $\widehat{\nu} = 0$ implies $\nu = 0$ without an extra uniqueness assumption. Convolve ν with a centred Gaussian approximate identity. Its value at every point is zero by that same Fourier integral. For every compactly supported continuous test function, its convolution with these Gaussians converges uniformly to the test function; integration against ν then proves that ν annihilates every such function.

Let μ be any probability measure of finite energy in (EIQ1), and write $\mu_*(dx) = \rho_{\alpha,\beta}(x) dx$ and $\nu = \mu - \mu_*$. The integrability needed for (EIQ21) holds. The $\mu \otimes \mu$ term has it by finite energy and the already established logarithmic moment; the $\mu_* \otimes \mu_*$ term has it by bounded compact support; and the cross term has it because the negative logarithmic singularity integrated against the bounded density is uniformly bounded, while the positive part is controlled by $\log(1+x) + \log(1+y)$. Expanding the energy therefore yields exactly

$$\mathcal{E}_{\alpha,\beta}(\mu) - \mathcal{E}_{\alpha,\beta}(\mu_*) = \int (U(x) - \ell) d\mu(x) - \iint \log|x - y| d\nu(x) d\nu(y) \geq 0. \quad (\text{EIQ23})$$

The first term is nonnegative by (EIQ17)–(EIQ19), and the second is strictly positive unless $\mu = \mu_*$. Infinite energy competitors cannot improve this finite value. Thus (EIQ11) is the unique global minimizing probability measure, with exactly the support $[a, b]$.

13.4 An explicit endpoint integral for the logarithmic moment

Retain m, R_s, u, v as defined above, and define

$$C = \frac{(u+v)^2}{4}, \quad \eta = \frac{v-u}{v+u}, \quad \zeta_s = \frac{b-a}{2(m+s+R_s)} \quad (s \geq 0). \quad (\text{EIQ24})$$

Thus $0 < \zeta_s \leq \eta < 1$ and $\zeta_0 = \eta$. For every $s \geq 0$ put

$$L_s = (m+s-R_s) \log C - (m-R_0) - 2R_s \log(1-\eta\zeta_s). \quad (\text{EIQ25})$$

Then the logarithmic moment of the equilibrium measure is exactly

$$\mathcal{L}(\alpha, \beta) := \int_a^b \log x \rho_{\alpha, \beta}(x) dx = \frac{\beta}{4\pi} \int_0^\infty \frac{L_0 - L_s}{\sqrt{s} R_s} ds. \quad (\text{EIQ26})$$

This is a convergent one-dimensional endpoint integral of explicitly given real elementary functions; its only endpoints are those specified uniquely in (EIQ3)–(EIQ5).

Here is a direct derivation, including the constants in (EIQ25). Set $h_0 = (b - a)/2$ and substitute $t = m + h_0 \cos \theta$ in arcsine integrals. The identities

$$t = C(1 + 2\eta \cos \theta + \eta^2), \quad t + s = \frac{m + s + R_s}{2}(1 + 2\zeta_s \cos \theta + \zeta_s^2)$$

give the absolutely convergent Fourier series

$$\log t = \log C + 2 \sum_{j \geq 1} \frac{(-1)^{j+1} \eta^j}{j} \cos(j\theta), \quad \frac{1}{t + s} = \frac{1}{R_s} \left(1 + 2 \sum_{j \geq 1} (-\zeta_s)^j \cos(j\theta) \right).$$

Orthogonality on $[0, \pi]$ gives

$$\begin{aligned} \frac{1}{\pi} \int_a^b \frac{\log t}{\sqrt{(b-t)(t-a)}} dt &= \log C, \\ \frac{1}{\pi} \int_a^b \frac{t \log t}{\sqrt{(b-t)(t-a)}} dt &= m \log C + (m - R_0), \\ \frac{1}{\pi} \int_a^b \frac{\log t}{(t + s) \sqrt{(b-t)(t-a)}} dt &= \frac{\log C + 2 \log(1 - \eta \zeta_s)}{R_s}. \end{aligned} \quad (\text{EIQ27})$$

The second identity uses $h_0 \eta = m - R_0$. Polynomial division

$$\frac{(b-t)(t-a)}{t+s} = -t + 2m + s - \frac{R_s^2}{t+s}$$

now proves, exactly,

$$L_s = \frac{1}{\pi} \int_a^b \frac{\log t \sqrt{(b-t)(t-a)}}{t+s} dt. \quad (\text{EIQ28})$$

Using the positive integral expression for h in (EIQ10), and the identity $1/[t(t+s)] = (1/t - 1/(t+s))/s$, yields (EIQ26). The exchanges are absolutely convergent: $\log t$ is bounded on $[a, b]$, $L_0 - L_s = O(s)$ at zero by (EIQ28), and $L_s = O(s^{-1})$ at infinity. Thus the integrand in (EIQ26) is $O(s^{1/2})$ at zero and $O(s^{-3/2})$ at infinity.

13.5 Parameter continuity and the variational derivative

The density and its logarithmic moment depend smoothly on $(\alpha, \beta) \in (0, \infty)^2$ in the following precise sense. On a compact parameter neighbourhood, (EIQ3)–(EIQ6) put a, b in a fixed compact subset of $(0, \infty)$, with $b - a$ bounded below by a positive number. The integral for h and each of its parameter and x derivatives converge uniformly for x in this fixed compact set: their absolute integrands are bounded by constants times $s^{1/2}$ near zero and $s^{-3/2}$ near infinity. Parameter derivatives of the endpoints are bounded on a smaller compact neighbourhood. Under the fixed-interval coordinate $x = a + (b - a)y$, $0 \leq y \leq 1$, the measure is exactly

$$\rho_{\alpha, \beta}(x) dx = \frac{(b - a)^2}{2\pi} \sqrt{y(1 - y)} h(a + (b - a)y) dy. \quad (\text{EIQ29})$$

All parameter derivatives of its coefficient are bounded by a constant times $\sqrt{y(1-y)}$. The same is true after multiplication by $\log(a + (b-a)y)$ or by $\sqrt{a + (b-a)y}$. Differentiation under this integral proves smoothness, in particular continuity, of both moments on the entire positive parameter quadrant.

Define $E_{\min}(\alpha, \beta) = \mathcal{E}_{\alpha, \beta}(\mu_{\alpha, \beta})$, where the measure is the unique minimizer proved above. Testing the minimizer for a parameter pair against the neighbouring potential, and then reversing the two tests, gives for $h > 0$

$$-\mathcal{L}(\alpha + h, \beta) \leq \frac{E_{\min}(\alpha + h, \beta) - E_{\min}(\alpha, \beta)}{h} \leq -\mathcal{L}(\alpha, \beta).$$

The corresponding inequalities for $h < 0$ and the proved continuity therefore give

$$\partial_{\alpha} E_{\min}(\alpha, \beta) = -\mathcal{L}(\alpha, \beta), \quad \partial_{\beta} E_{\min}(\alpha, \beta) = \int \sqrt{x} d\mu_{\alpha, \beta}(x). \quad (\text{EIQ30})$$

No unproved differentiability of a minimizing selection is being used. For the opposite convention $F(\alpha, \beta) = \sup_{\mu} \{ \iint \log |x - y| d\mu(x) d\mu(y) - \int V_{\alpha, \beta} d\mu \} = -E_{\min}(\alpha, \beta)$, the exact derivative is therefore $\partial_{\alpha} F(\alpha, \beta) = \mathcal{L}(\alpha, \beta)$.

The second moment in (EIQ30) has an exact value. Push the minimizing measure forward by the explicit dilation $x \mapsto c^2 x$, $c > 0$. Writing $M = \int \sqrt{x} d\mu_{\alpha, \beta}(x)$, its energy minus the energy at $c = 1$ is precisely $\beta(c-1)M - 2(\alpha+1) \log c$. Its derivative at the minimizing value $c = 1$ is zero. Hence

$$\int \sqrt{x} d\mu_{\alpha, \beta}(x) = \frac{2(\alpha+1)}{\beta}. \quad (\text{EIQ31})$$

The same explicit dilation, or (EIQ3)–(EIQ5), proves that the map $x \mapsto (\pi/\beta)^2 x$ pushes $\mu_{\alpha, \pi}$ to $\mu_{\alpha, \beta}$, including the density and its two endpoints. Consequently

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\alpha, \pi) + 2 \log(\pi/\beta). \quad (\text{EIQ32})$$

For the particular original limiting potential $V(x) = \pi\sqrt{x} - 2 \log x$, the substitutions are exactly $\alpha = 2$, $\beta = \pi$. Equations (EIQ5), (EIQ10), and (EIQ26) become

$$uK(k) = 2, \quad vE(k) = 4, \quad h(x) = \frac{1}{2x} \int_0^{\infty} \frac{\sqrt{s}}{(x+s)R_s} ds, \quad \mathcal{L}(2, \pi) = \frac{1}{4} \int_0^{\infty} \frac{L_0 - L_s}{\sqrt{s}R_s} ds. \quad (\text{EIQ33})$$

In particular the preceding proof includes positivity, unit mass, the full Euler inequality, the unique global minimizer, and the parameter continuity needed at these actual parameters.

14 The exact Gamma ensemble and the leading growing return

GEL1. Source, exponents, and the literal coordinate map. The source throughout is

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \sigma(\mathbb{R}) = \sqrt{2\pi}, \quad d\nu(t) = \frac{|\Gamma(1/4 + i\sqrt{t}/2)|^2}{2\pi\sqrt{t}} dt. \quad (\text{GEL1})$$

Here ν is the full pushforward under $y \mapsto y^2$; both half-axes are present in its density. For integers $a, n \geq 0$, retain

$$H_n^{(a)} = \det \left[\int_0^\infty t^{a+i+j} d\nu(t) \right]_{i,j < n}.$$

The definition extends to real $a > -1/2$; all its integrals converge. For $n \geq 1$, expansion of the two Vandermonde determinants followed by integration term by term gives the exact identity

$$H_n^{(a)} = \frac{1}{n!} \int_{(0,\infty)^n} \prod_{i < j} (t_i - t_j)^2 \prod_{i=1}^n t_i^a d\nu(t_i). \quad (\text{GEL2})$$

Indeed expansion gives $n!$ copies of the Gram determinant after one of the two permutation indices is fixed. Absolute integrability follows by expanding the squared polynomial and using the existing moments.

Fix a positive real coordinate scale q , independently of a, n , and make the literal map $t_i = q^2 x_i$. Introduce the reference probability measure

$$d\omega(x) = \frac{1}{2} x^{-1/2} e^{-\sqrt{x}} dx, \quad B_q(x) = 2qe^{\sqrt{x}} \sigma(q\sqrt{x}).$$

Its mass is one by $x = u^2$. This reference measure does not replace or rescale the source: the exact Radon–Nikodym identity is $d\nu(q^2 x) = B_q(x) d\omega(x)$, including its whole mass. Consequently

$$H_n^{(a)} = q^{2an+2n(n-1)} J_{n,q}(a), \quad J_{n,q}(a) = \frac{1}{n!} \int \prod_{i < j} (x_i - x_j)^2 \prod_i x_i^a B_q(x_i) d\omega(x_i). \quad (\text{GEL3})$$

The powers in this identity come separately from the n factors t_i^a and the $n(n-1)/2$ squared differences. No determinant, source mass, or Jacobian has been omitted.

GEL2. An elementary global Gamma bound with the exact exponential. Put $\lambda = 1/4, b \geq 0$, and $C_\Gamma = \Gamma(1/4)^2/(2\pi)$. Euler's finite limit gives

$$\frac{|\Gamma(\lambda + ib)|^2}{\Gamma(\lambda)^2} = \prod_{j=0}^{\infty} \left(1 + \frac{b^2}{(j + \lambda)^2} \right)^{-1}.$$

For clarity, the finite limit follows by applying the beta integral to $\int_0^1 t^{z-1} (1-t)^N dt = N!/[z(z+1) \cdots (z+N)]$, replacing t by u/N , and using dominated convergence for $\text{Re } z > 0$. Taking squared moduli and dividing by the same limit at $z = \lambda$ proves the displayed product. Its logarithm converges absolutely because its summand is $O((j+1)^{-2})$.

The decreasing function $v \mapsto \log(1 + b^2/v^2)$ gives, with $S = \sum_{j \geq 0} \log(1 + b^2/(j + \lambda)^2)$,

$$I \leq S \leq I + \log(1 + b^2/\lambda^2), \quad I = \int_\lambda^\infty \log(1 + b^2/v^2) dv = \pi b - \lambda \log(1 + b^2/\lambda^2) - 2b \arctan(\lambda/b).$$

The integral is evaluated by differentiating $v \log(1 + b^2/v^2) + 2b \arctan(v/b)$; at $b = 0$ it is zero by continuity. Since $0 \leq 2b \arctan(\lambda/b) \leq 2\lambda$, substitution of $b = |y|/2$ proves the global, finite inequalities

$$C_\Gamma (1 + 4y^2)^{-3/4} e^{-\pi|y|/2} \leq \sigma(y) \leq C_\Gamma e^{1/2} (1 + 4y^2)^{1/4} e^{-\pi|y|/2}. \quad (\text{GEL4})$$

Both inequalities include $y = 0$. In particular, for $C_0 = \Gamma(1/4)^2/\pi$,

$$B_q(x) \leq C_0 q e^{1/2} (1 + 2q)^{1/2} e^{-(\pi q/2 - 3/2)\sqrt{x}} \quad (x > 0). \quad (\text{GEL5})$$

Here $(1 + 4q^2 x)^{1/4} \leq (1 + 2q\sqrt{x})^{1/2} \leq (1 + 2q)^{1/2} (1 + \sqrt{x})^{1/2} \leq (1 + 2q)^{1/2} e^{\sqrt{x}/2}$. On each fixed compact interval $[A, B] \subset (0, \infty)$, the lower bound is

$$B_q(x) \geq C_0 q (1 + 4q^2 B)^{-3/4} e^{-\pi q \sqrt{x}/2}. \quad (\text{GEL6})$$

Thus the exact exponential, and the size and position of the remaining source factor, have been proved directly from the original Gamma function.

GEL3. The energy object and its original parameter domain. For $\alpha, \beta > 0$ put

$$V_{\alpha, \beta}(x) = \beta \sqrt{x} - \alpha \log x, \quad K_{\alpha, \beta}(x, y) = \log |x - y| - \frac{1}{2} V_{\alpha, \beta}(x) - \frac{1}{2} V_{\alpha, \beta}(y),$$

$$\mathcal{F}(\alpha, \beta) = \sup_{\mu \in \mathcal{P}((0, \infty))} \iint K_{\alpha, \beta}(x, y) d\mu(x) d\mu(y). \quad (\text{GEL7})$$

The kernel is bounded above, so its integral has an unambiguous value in $\mathbb{R} \cup \{-\infty\}$ for every probability in this domain. Whenever its value is finite it equals $\iint \log |x - y| d\mu^2 - \int V_{\alpha, \beta} d\mu$. Indeed $K \leq r(x) + r(y)$, with $r = \log(1 + x) - V_{\alpha, \beta}/2$ bounded above and tending to $-\infty$ at both endpoints; finite kernel integral therefore forces the $\sqrt{x}$ and $|\log x|$ moments to be finite. The positive logarithmic part is then integrable, and finite kernel integral forces the negative part to be integrable as well. This defines the extended energy without forming an indeterminate difference of infinite quantities. The equilibrium calculation supplied with this section proves existence, uniqueness, and the full exterior variational inequality for its density $\rho_{\alpha, \beta}$, with support $[u^2, v^2]$. Its exact endpoints and positive density are

$$uK(\kappa) = \frac{\alpha\pi}{\beta}, \quad vE(\kappa) = \frac{(\alpha + 2)\pi}{\beta}, \quad \kappa^2 = 1 - u^2/v^2, \\ \rho_{\alpha, \beta}(x) = \frac{\beta \sqrt{(v^2 - x)(x - u^2)}}{4\pi^2 x} \int_0^\infty \frac{\sqrt{s} ds}{(x + s) \sqrt{(u^2 + s)(v^2 + s)}} \quad (u^2 < x < v^2). \quad (\text{GEL8})$$

The endpoint proof shows continuous parameter dependence, finite entropy relative to ω , finite logarithmic energy, and

$$\partial_\alpha \mathcal{F}(\alpha, \beta) = \int \log x \rho_{\alpha, \beta}(x) dx. \quad (\text{GEL9})$$

In particular $\mathcal{F}$ and this derivative are continuous on the positive parameter quadrant. The following ensemble argument proves the connection to GEL3; it does not presume a random-matrix asymptotic theorem.

GEL4. Complete free-energy limit from a compactified kernel. Let $n \rightarrow \infty$, $q/n \rightarrow \theta > 0$, and let $\alpha > 0$ be fixed. Then the actual integral in GEL3 satisfies

$$\boxed{\frac{1}{n^2} \log J_{n, q}(n\alpha) \longrightarrow \mathcal{F}(\alpha, \pi\theta/2).} \quad (\text{GEL10})$$

Here is a proof of both bounds. For the upper bound GEL5 removes the factor $[C_0 q e^{1/2} (1 + 2q)^{1/2}]^n / n!$, whose logarithm divided by n^2 tends to zero. The remaining exponent is

$$2 \sum_{i < j} \log |x_i - x_j| - n \sum_i V_{\alpha, \beta_n}(x_i), \quad \beta_n = \frac{\pi q}{2n} - \frac{3}{2n}.$$

For $n > 1$ it is exactly $\sum_{i \neq j} K_{Q_n}(x_i, x_j)$, where

$$Q_n = \frac{n}{n-1} V_{\alpha, \beta_n}, \quad K_Q(x, y) = \log |x - y| - \frac{1}{2} Q(x) - \frac{1}{2} Q(y).$$

Write $\beta = \pi\theta/2$. Choose $\varepsilon > 0$ so that $\alpha - \varepsilon > 0$ and $\beta - 3\varepsilon > 0$. For all sufficiently large n , the two coefficients of Q_n differ from (α, β) by at most ε . It follows pointwise that $Q_n \geq Q_- := V_{\alpha-\varepsilon, \beta-3\varepsilon}$. Indeed its difference has the form $b\sqrt{x} - a \log x$, where $b \geq 2\varepsilon$ and $0 \leq a \leq 2\varepsilon$; it is nonnegative for $x < 1$, and for $x \geq 1$ use $\log x \leq \sqrt{x}$.

Compactify $(0, \infty)$ by adjoining its two endpoints. The function K_{Q_-} , assigned value $-\infty$ on the diagonal and at either added endpoint, has continuous truncations $K_M = \max\{K_{Q_-}, -M\}$. To verify the endpoint assertion uniformly in the other variable, use

$$K_{Q_-}(x, y) \leq r(x) + r(y), \quad r(x) = \log(1+x) - Q_-(x)/2.$$

The function r is bounded above and tends to $-\infty$ at both endpoints. Continuity at a diagonal point in the interior follows from $\log |x-y| \rightarrow -\infty$ and local boundedness of Q_- . For the empirical probability $L_n = n^{-1} \sum_i \delta_{x_i}$,

$$\frac{1}{n^2} \sum_{i \neq j} K_{Q_-}(x_i, x_j) \leq \iint K_M dL_n^2 + \frac{M}{n} \leq s_M + \frac{M}{n}, \quad s_M = \max_{\mu} \iint K_M d\mu^2.$$

Since ω has mass one, integration preserves this upper bound. Moreover $s_M \downarrow \mathcal{F}(\alpha - \varepsilon, \beta - 3\varepsilon)$. For completeness, choose maximizing probabilities μ_M . Compactness gives a convergent subsequence. For every fixed L , the inequality $K_M \leq K_L$ when $M \geq L$, followed by continuity of the integral of K_L , bounds the subsequential limit of s_M by $\iint K_L d\mu^2$. Decrease L 's truncation to the untruncated kernel. Monotone convergence after subtracting a common upper bound gives $\lim s_M \leq \iint K_{Q_-} d\mu^2 \leq \mathcal{F}$. The reverse inequality follows by testing every K_M on an arbitrary probability of finite energy. Probabilities charging an added endpoint have energy $-\infty$, so they do not alter the supremum. First take $n \rightarrow \infty$, then $M \rightarrow \infty$, and finally $\varepsilon \downarrow 0$. Continuity proved in GEL9 yields the upper bound in GEL10.

For the lower bound choose exactly $d\mu(x) = \rho_{\alpha, \beta}(x)dx$, supported on $[A, B]$, and use GEL6. Write $p = d\mu/d\omega$. Restricting the integral to this support and changing measure to μ^n , Jensen's inequality gives

$$\begin{aligned} \log J_{n,q}(n\alpha) &\geq -\log n! + n \log\{C_0 q(1 + 4q^2 B)^{-3/4}\} \\ &\quad + n(n-1) \iint \log |x - y| d\mu(x) d\mu(y) \\ &\quad - n^2 \int V_{\alpha, \pi q/(2n)} d\mu - n \int \log p d\mu. \end{aligned}$$

All logarithms on the right are integrable: the equilibrium density is bounded, has square-root decay at its finite positive endpoints, and ω has a positive smooth density there. The prefactor contributes $O(n \log q + n \log n)$, and the entropy contributes $O(n)$. Division by n^2 therefore yields exactly the value of GEL7 at its maximizer. This proves GEL10.

GEL5. Finite positive shifts by a proved convex squeeze. For each fixed n, q , differentiation under the integral in GEL3 on every compact subinterval of $a > -1/2$ gives

$$\frac{d}{da} \log J_{n,q}(a) = \mathbb{E}_a \sum_i \log x_i, \quad \frac{d^2}{da^2} \log J_{n,q}(a) = \text{Var}_a \left(\sum_i \log x_i \right) \geq 0. \quad (\text{GEL11})$$

The expectation is with respect to the entire positive integrand in GEL3 divided by its integral. Extra logarithmic factors are integrable at zero because $a > -1/2$, and at infinity by GEL4. Hence the functions $F_{n,q}(z) = n^{-2} \log J_{n,q}(nz)$ are differentiable and convex. Suppose now

$$q/n \rightarrow 2, \quad a/n \rightarrow 2, \quad l > 0, \quad l/n \rightarrow 0.$$

Set

$$\mathcal{L} = \int_{u^2}^{v^2} \log x \rho_{2,\pi}(x) dx, \quad uK(\kappa) = 2, \quad vE(\kappa) = 4. \quad (\text{GEL12})$$

For fixed $0 < \eta < 2$, both endpoints of the secant from a/n to $(a+l)/n$ lie in $(2 - \eta/2, 2 + \eta/2)$ for large n . Convexity bounds that secant from below by the secant from $2 - \eta$ to $2 - \eta/2$, and from above by the secant from $2 + \eta/2$ to $2 + \eta$. Apply GEL10 at these four fixed positive arguments, then let $\eta \downarrow 0$ and use GEL9. This proves

$$\frac{\log J_{n,q}(a+l) - \log J_{n,q}(a)}{nl} \rightarrow \mathcal{L}.$$

Combining this with the exact source factors in GEL3 gives

$$\log \frac{H_n^{(a+l)}}{H_n^{(a)}} = 2nl \log q + nl\mathcal{L} + o(nl). \quad (\text{GEL13})$$

This proof covers moving starting exponents a ; it never differentiates the error in a free-energy asymptotic. The remainder assertion has the precise scope $o(nl)$ along the displayed sequences.

GEL6. Both original matrices and the low moment. Return to the actual packet

$$k = 4l + 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2 = 2n, \quad m \geq 1 \text{ fixed}, \quad l \rightarrow \infty.$$

Retain the exact parity identity from HCT35, $Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2$. Use GEL13 at the four literal factors, with sizes $n, n+1, n, n$ and starting exponents $q, q, q+1, q+1$, respectively. Every sequence satisfies its stated domain. Addition gives

$$\log \frac{Z_{q+l}}{Z_q} = 2(2q+1)l \log q + (2q+1)l\mathcal{L} + o(ql). \quad (\text{GEL14})$$

Thus the contribution from the two original full matrices has been evaluated, with all four parity blocks still identifiable.

Write $\mu_a = \int y^{2a} d\sigma(y)$. GEL4 gives, for integer $a \geq 1$,

$$\log \mu_a = 2a \log a + O(a). \quad (\text{GEL15})$$

One elementary proof of the upper bound uses $(1 + 4y^2)^{1/4} \leq 1 + 2y$ for $y \geq 0$; its integrals are $(2a)!/(\pi/2)^{2a+1}$ and $2(2a+1)!/(\pi/2)^{2a+2}$, multiplied by the exact constant $2C_\Gamma e^{1/2}$. The bound $N! \leq N^N$ suffices. For the lower bound restrict the positive-axis integral to $[a, a+1]$: there $(1 + 4y^2)^{-3/4} \geq 5^{-3/4}(a+1)^{-3/2}$, $y^{2a} \geq a^{2a}$, and $e^{-\pi y/2} \geq e^{-\pi(a+1)/2}$. These prove both sides of GEL15 with constants independent of a .

The real-exponent function $a \mapsto \log \mu_a$ is convex by the same variance proof as GEL11. Since q is even and $1 \leq l \leq q$, its secant on $[q, q+l]$ lies between the secants on $[q/2, q]$ and $[2q, 4q]$. GEL15 bounds both latter secants in absolute value by $O(\log q)$. Therefore

$$\left| \log \frac{\mu_{q+l}}{\mu_q} \right| = O(l \log q) = o(ql). \quad (\text{GEL16})$$

The original low moment is precisely $\mu_{q+l}/\mu_q = m_{2q+2l}/m_{2q}$, so no index is reinterpreted.

GEL7. The leading value of the unchanged signed centre. Keep the source product and its exact row intervals:

$$P_k = - \sum_{a=q}^{2q-1} \sum_{j=0}^{2l-1} \log(a+j+1/2) - \sum_{a=q+1}^{2q} \sum_{j=0}^{2l-1} \log(a+j+1/2).$$

The two outer sums each have q terms. Since $l/q \rightarrow 0$, direct Riemann sums, with log Lipschitz on $[1, 3]$, give

$$P_k = -4ql \log q - 4ql \int_1^2 \log x \, dx + o(ql) = -4ql \log q - 4ql(2 \log 2 - 1) + o(ql). \quad (\text{GEL17})$$

Explicitly, omitting $(j + 1/2)/q$ changes each logarithm by at most $2l/q$, hence changes the total by at most $8l^2$; the ordinary left and right Riemann sums together have error $O(l)$.

The full finite product calculation WGP10 strengthens this leading identity to the following enclosure for the very same P_k :

$$P_k = -4lq \log q - 4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} + \mathcal{E}_{q,l}, \quad (\text{GEL17a})$$

$$- \left\{ \frac{l}{6q} + \frac{8l^3 + l}{12q^3} + \frac{8l^4 + 2l^2}{3q^2} \right\} \leq \mathcal{E}_{q,l} \leq \frac{l^2}{3q^2}.$$

Its proof uses the literal midpoint coordinates in WGP3, the signed quadrature remainders WGP4–5, and the nonnegative cubic remainder WGP6–9. The subsequent exact refinement WGP12–14 gives

$$P_k = -4lq \log q - 4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} - \frac{l^4 + l^2/4}{q^2} + \mathcal{E}_{q,l}^{(3)}, \quad (\text{GEL17b})$$

$$-\frac{l}{6q} - \frac{8l^3 + l}{12q^3} \leq \mathcal{E}_{q,l}^{(3)} \leq \frac{l^2}{3q^2} + \frac{8l^4 + 2l^2}{6q^4} + \frac{7(192l^5 + 80l^3 - 2l)}{1440q^3}.$$

These original endpoint contributions are retained when the finer q -scale calculation is continued. In particular the term $-4l^2 \log 2$ has order q at $m = 1$, and the cubic term tends to $-1/256$ there; the leading limit does not remove either term.

The original quantity is exactly $W_k = P_k - \log(\mu_{q+l}/\mu_q) + \log(Z_{q+l}/Z_q)$. Substitute GEL14, GEL16, and GEL17. The surviving $2l \log q$ is $o(ql)$, so the leading growing-degree calculation is

$$\frac{W_k}{ql} \longrightarrow \mathcal{C}_\Gamma := 2\mathcal{L} - 4(2 \log 2 - 1), \quad \mathcal{L} = \int_{u^2}^{v^2} \log x \, \rho_{2,\pi}(x) \, dx, \quad uK(\kappa) = 2, \, vE(\kappa) = 4.$$

(GEL18)

Every measure, exponent, matrix dimension, and source factor entering this limit has been connected to its original object by GEL1–3 and GEL14. The exact positive density and unique endpoint equations in GEL8 make the constant an explicitly defined one-dimensional equilibrium integral. This leading result has remainder $o(ql)$. It supplies no claim of an $o(q)$ remainder, and hence does not erase the next finite-scale calculation required by the original arithmetic receiver.

The fully proved original-polynomial recurrence estimate RWB20–27 also applies to this identical W_k . It gives the finite bounds $lq/64 \leq W_k \leq 6lq$, and its narrower finite lower remainder gives $\liminf W_k/(lq) \geq 4 \log(27/(8\pi))$. Combining those proved inequalities with the now proved existence of GEL18 yields

$$4 \log \frac{27}{8\pi} \leq \mathcal{C}_\Gamma \leq 6, \quad \mathcal{C}_\Gamma > 0.$$

(GEL18a)

The strict sign is analytic: RWB22 proves $4 \log(27/(8\pi)) > 104/365 > 0$ using rational integral bounds for π and the elementary logarithm inequality. Numerical evaluation of the equilibrium integral is unnecessary for this sign conclusion.

Finally LET20, whose error is $O(k) = o(q)$ at fixed original m, δ, γ , transfers the proved limit without changing its sign:

$$\frac{\mathfrak{T}_k}{ql} \longrightarrow \mathcal{C}_\Gamma, \quad \frac{R_k^0 - R_k^\sigma}{ql} \longrightarrow -\mathcal{C}_\Gamma. \quad (\text{GEL19})$$

The arithmetic identity remains $B_{\text{ar}} = B_0 + \delta_\sigma - \mathfrak{T}_k$, including its original δ_σ allowance. Thus GEL19 evaluates the Gamma summand and its exact signed map into that identity; it makes no unsupported growth assertion about either of the other two summands.

15 The evaluated Gamma term in the original arithmetic and cohomology receiver

Retain the actual packet $k = 4l + 1$, $e = 1 + k(m - 1)$, $q = e(k + 1)^2$, $c = k/2$, $0 < \delta < 1/2$, $\gamma > 2$, and the original polynomials

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \quad f(S) = \prod_{j=0}^{l-1} (S - c - 2j - 1/2). \quad (\text{WGR1})$$

Here $E = \mathbb{C}[S]/(\chi)$, $J_N : \mathbb{C}[S]_{\leq N} \rightarrow E$ is the literal remainder map, and the source line remains $S = c + iy$. The parameters, primary multiplicities, and polynomial maps have not been changed by the evaluation of W_k .

Let $\mathfrak{r}_\chi$ be the original scalar norm ratio at $N = q$, and let $\mathfrak{R}_{2q-1}, \mathfrak{R}_{2q}$ be the two original relation-Gram ratios from JSR12. Write the exact scalar errors

$$\begin{aligned} e_0 &= \log \mathfrak{r}_\chi - \log(m_{2q+2l}/m_{2q}), \\ e_r &= \log \mathfrak{R}_{q+r-1} - \log \frac{\det \mathbf{M}_{q+l,r}}{\det \mathbf{M}_{q,r}}, \quad r = q, q+1, \\ (\mathbf{M}_{a,r})_{ij} &= \int_{\mathbb{R}} y^{2a+i+j} \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad 0 \leq i, j < r. \end{aligned} \quad (\text{WGR2})$$

Thus the exponent index m_j is the literal power j , and both matrices use the full original mass $\sqrt{2\pi}$. The exact coefficient map $P(S) \mapsto P(c + iy)$ has entries $J_{hj} = \binom{j}{h} c^{j-h} i^h$ and determinant of absolute value one. In particular the high reference Gram is its congruence $J^* \mathbf{M}_{a,r} J$, not a different-coordinate determinant with an unrecorded power of q . The bulk multiplication $\chi(c + iy)/(iy)^q$ and the full positive inner and far forms are the exact source maps and corrections proved in CTR and LET.

On the explicit domain $k \geq 2048\sqrt{\delta^2 + \gamma^2}$, LET18 and CTR48 prove

$$e_0 \in [-a_0, b_0], \quad e_q + e_{q+1} \in [-A, B], \quad A = 2lE_0 + 2(L_q + L_{q+1}), \quad B = A + (2q+1)\Delta_f, \quad (\text{WGR3})$$

where $a_0 = E_0 + 2L_1$, $b_0 = a_0 + \Delta_f$, and every constant E_0, Δ_f, L_r is the finite original integral constant printed in LET7-16 and BHR2. In particular $A, B = O(k)$ for fixed $m = 1$ and $O(1)$ for fixed $m \geq 2$; $a_0, b_0 = O(1)$ and $O(k^{-1})$ in those respective branches. These statements were proved on the full source, including the low rank one, and are used within their stated domain here.

The exact identity LET19, retaining the low minus sign, is

$$\mathfrak{T}_k = W_k - e_0 + e_q + e_{q+1}, \quad \mathcal{H}_k := R_k^0 - R_k^\sigma = -\mathfrak{T}_k. \quad (\text{WGR4})$$

RWB20-27 now supplies an evaluated pair of finite numbers

$$\begin{aligned} w_{q,l}^- &= \max \left\{ \frac{lq}{64}, 4lq \log \frac{27}{8\pi} - 4l^2 - 6l \log(3/2) - 3l \right. \\ &\quad \left. - \frac{9l}{4q-1} - \frac{2l(l+1/4)}{q} \right\}, \quad w_{q,l}^+ = 6lq, \quad w_{q,l}^- \leq W_k \leq w_{q,l}^+. \end{aligned} \quad (\text{WGR5})$$

This assertion holds at every original $l, m \geq 1$; the extra threshold in WGR3 is required only when returning to the original χ source. Substituting WGR5 and then the three signed errors into WGR4 proves

$$\boxed{-w_{q,l}^+ - B - a_0 \leq \mathcal{H}_k \leq -w_{q,l}^- + A + b_0.} \quad (\text{WGR6})$$

Indeed $\mathcal{H}_k = -W_k + e_0 - e_q - e_{q+1}$. This proves both signs separately; no positive high error has been assigned the sign of the negative low endpoint.

The complete retained arithmetic and mixed-state map. Use exactly the previous signed arithmetic difference δ_k^σ and the full baseline $\mathcal{B}_k^0$ from JSR16–21. For either original source, the quotient, kernel and boundary forms are

$$G_N = (J_N H_N^{-1} J_N^*)^{-1}, \quad H_N^K = I_K^* G_N I_K, \quad Q_N^B = (\Lambda G_N^{-1} \Lambda^*)^{-1}. \quad (\text{WGR7})$$

Here $\Lambda : E \rightarrow B_{\text{obs}} = \text{im } \Lambda$ is the original boundary observation, $K = \ker \Lambda$, and $I_K : K \hookrightarrow E$ is the original inclusion. The chosen original lift $S_* : B_{\text{obs}} \rightarrow E$ satisfies $\Lambda S_* = I_{B_{\text{obs}}}$. Thus $[I_K, S_*] : K \oplus B_{\text{obs}} \rightarrow E$ is the exact coefficient isomorphism. The exact sequence is represented in this common coefficient frame $[I_K, S_*]$. Orthogonal projection subtracts a kernel column from each lift in S_* and is a triangular change with determinant one, proving

$$\det H_N^K \det Q_N^B = |\det[I_K, S_*]|^2 \det G_N. \quad (\text{WGR8})$$

The same frame is used for both sources, so its determinant cancels in their ratio. The source comparison gives the original nonnegative deficits $b_N^K + b_N^B = b_N \leq \Xi_{k,N}$ and the four signed sums

$$\begin{aligned} X_k &= -b_{q-1} - b_q + b_{2q-1} + b_{2q}, \\ a_k^- &= \Xi_{k,q-1} + \Xi_{k,q}, \quad a_k^+ = \Xi_{k,2q-1} + \Xi_{k,2q}, \quad -a_k^- \leq X_k \leq a_k^+. \end{aligned} \quad (\text{WGR9})$$

The same formula applies to $\Delta F_K, \Delta F_B$ by using the corresponding superscripts, and to δ_k^σ by using the full quotient. The last inequality follows by dropping the two positive terms for the lower bound and the two negative terms for the upper bound. Thus the original mixed kernel and boundary directions remain in the exact map, rather than being lost in a scalar positivity assertion. The previous finite source estimates $a_k^- \leq D_k^-$ and $a_k^+ \leq D_k^+$, with all original constants in JSR17–18, continue to apply separately.

The original arithmetic equality and the evaluated finite bounds are therefore

$$\begin{aligned} \mathcal{B}_k^{\text{ar}} &= \mathcal{B}_k^0 + \mathcal{H}_k + \delta_k^\sigma, & \Delta_k^\Gamma &= \mathcal{H}_k + \delta_k^\sigma, \\ \mathcal{B}_k^0 - w_{q,l}^+ - B - a_0 - a_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 - w_{q,l}^- + A + b_0 + a_k^+. \end{aligned} \quad (\text{WGR10})$$

Add WGR9 to WGR6 to prove WGR10. Replacing $a_k^\pm$ by their respective outer bounds $D_k^\pm$ gives the corresponding fully estimated arithmetic interval. In particular the original nonnegative HC residual has the finite enclosure

$$\begin{aligned} \max\{0, \mathcal{B}_k^0 - w_{q,l}^+ - B - a_0 - D_k^- - 4q \log(D_h k)\} &\leq \mathcal{R}_k := \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k) \\ &\leq \mathcal{B}_k^0 - w_{q,l}^- + A + b_0 + D_k^+ - 4q \log(D_h k). \end{aligned} \quad (\text{WGR11})$$

Its constant D_h and its threshold are the previous original ones. For every finite exact W_k , WGR6 contains the sharper WRC5 interval. Consequently it provides an evaluated bound with the original source constants on the growing family; it does not claim to narrow an interval already using the exact scalar W_k as input.

A definite growing arithmetic contribution. Fix an original packet (δ, γ, m) and let $k = 4l + 1 \rightarrow \infty$. The already proved JSR18–21 finite source estimates give

$$\begin{cases} \delta_k^\sigma / (kq) = O_h((\log k/k)^{5/7}), & m = 1, \\ \liminf \frac{\delta_k^\sigma}{q \log k} \geq -(64m + 245), \quad \limsup \frac{\delta_k^\sigma}{q \log k} \leq 128m + 490, & m \geq 2. \end{cases} \quad (\text{WGR12})$$

Thus $\delta_k^\sigma / (kq) \rightarrow 0$ in both branches. In the second branch multiply the finite upper and lower constants by $\log k/k \rightarrow 0$; in the first branch the displayed rate tends to zero. Also WGR3 implies $(e_0 - e_q - e_{q+1}) / (kq) \rightarrow 0$, retaining its original asymmetric bounds. Since $l/k = (k-1)/(4k) \rightarrow 1/4$, RWB23 and RWB27 yield

$$\boxed{-\frac{3}{2} \leq \liminf \frac{\Delta_k^\Gamma}{kq} \leq \limsup \frac{\Delta_k^\Gamma}{kq} \leq -\log \frac{27}{8\pi} < 0.} \quad (\text{WGR13})$$

This uses $\Delta_k^\Gamma = -W_k + e_0 - e_q - e_{q+1} + \delta_k^\sigma$ and the proved finite bounds, so it does not infer a limit from the exploratory numerical recurrence. In particular the Gamma/arithmetic difference is strictly negative for all sufficiently large original degrees. It remains a term in the full arithmetic equality WGR10.

The preceding full-source equilibrium calculation GEL18–19 evaluates this term more precisely, proving

$$\boxed{\frac{\Delta_k^\Gamma}{kq} \longrightarrow -\frac{\mathcal{C}_\Gamma}{4}, \quad 4 \log \frac{27}{8\pi} \leq \mathcal{C}_\Gamma \leq 6.} \quad (\text{WGR14})$$

Indeed GEL18 proves $W_k/(lq) \rightarrow \mathcal{C}_\Gamma$, while the two other summands divided by kq vanish by WGR12 and WGR3. The finite RWB bounds give the stated interval for the exact equilibrium integral $\mathcal{C}_\Gamma$. Combining this result with WGR10–11 gives the evaluated relation to the original baseline and residual:

$$\frac{\mathcal{B}_k^0 - \mathcal{B}_k^{\text{ar}}}{kq} \longrightarrow \frac{\mathcal{C}_\Gamma}{4}, \quad \frac{\mathcal{R}_k - \mathcal{B}_k^0}{kq} \longrightarrow -\frac{\mathcal{C}_\Gamma}{4}, \quad \liminf \frac{\mathcal{B}_k^0}{kq} \geq \frac{\mathcal{C}_\Gamma}{4}. \quad (\text{WGR15})$$

The second assertion also uses $4 \log(D_h k)/k \rightarrow 0$ at fixed $D_h > 0$; the third follows from the independent original inequality $\mathcal{R}_k \geq 0$. These are evaluated differences and a necessary baseline bound with the original arithmetic object retained. They do not assert that the baseline or the nonnegative residual has vanished. The finite q -scale terms in WGP13 and the finite allowances in WGR10–12 remain available for the next calculation on those actual objects. In particular the $o(lq)$ remainder in GEL18 has not been replaced by an unproved $o(q)$ remainder.